

Review

Integrative approaches to enhance reproductive resilience of crops for climate-proof agriculture

RECROP COST⁺

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ABSTRACT

Worldwide agricultural systems are threatened by rising temperatures, extreme weather events, and shifting climate zones. Climate change-driven failure in sexual reproduction is a major cause for yield reduction in horticultural and grain crops. Consequently, understanding how climate change affects reproductive processes in crops is crucial for global food security and prosperity. The development of climate-proof crops, including maize, wheat, barley, rice, and tomato, requires new genetic material and novel management practices to ensure high productivity under less favorable conditions. Safeguarding successful plant reproduction is challenging due to the complex nature of this biological process, and therefore, a multifaceted approach is the key to success. In this review, we provide an overview of the processes underlying plant reproduction and how they are affected by different abiotic stresses related to climate change. We discuss how genetics, advanced breeding technologies, biotechnological innovations, and sustainable agronomic practices can collectively contribute to the development of resilient crop varieties. We also highlight the potential of artificial intelligence (AI) in optimizing breeding strategies, predicting climate impacts, and improving crop management practices to enhance reproductive resilience and ensure food security. Lastly, we discuss the vision of a new era in agriculture where diverse actors and stakeholders cooperate to create climate-proof crops.

1. Introduction

The world's agricultural systems face significant challenges due to the ongoing effects of climate change and growing global demand for food production. An increase in global mean temperature of more than 1 °C and strong thermal anomalies during the period 2011–2023 have been observed, and associated with a higher frequency and intensity of extreme weather events such as heat waves, cold or drought periods, as well as heavy rainfalls causing floods (Forster et al., 2024). These phenomena have already caused disasters and have dire consequences for all actors along the food supply chain (Schmidt and Felsche, 2024).

Climate change impacts agricultural yields globally in varied ways. For maize, for example, it was estimated that each degree-Celsius increase in air temperature will reduce global yield by 7.4% leading to a 45% reduction by 2080 (Deryng et al., 2014; Zhao et al., 2017). In India, warming temperatures reduced wheat grain yields by 5.2% from 1981 to 2009 (Gupta et al., 2017). Since 1979, Europe experienced reduced average production-weighted continent-wide wheat and barley yields by 2.5% and 3.8%, respectively (Faye et al., 2023), while grain maize yield decreased by 4% during the past 14 years (The State of Food Security and Nutrition in the World, 2022). Moreover, short-term temperature and rainfall fluctuations also impact productivity. From 2010 to 2012, higher temperatures and lower precipitation in the Southern U.S., Western Russia, Western Australia, and East Africa caused a 14–80% grain yield decline in crops such as barley and wheat, leading to famine in East Africa (Anderson et al., 2020). For processing tomatoes, a decline by 6% in yield is predicted till 2050 in certain regions (Cammarano et al., 2022). In addition, climate zones are shifting poleward, changing the distribution of productive agricultural areas (Parry et al., 2005). In the future, warmer temperatures will extend growing seasons and open new areas for maize and wheat cultivation, among other crops, in Northern Europe, Russia and Canada, while conditions will become restrictive for important crops such as maize and wheat in other regions (Franke et al., 2022). Notably, many developed countries in cooler climates are likely to benefit from increased yields in some crops, while the majority of developing countries in subtropics and tropics will likely face reduced yields, widening the wealth gap (Anderson et al., 2020).

Considering the expected increase of 35–56% in global food demand by 2050, the current situation begs for adopting new strategies that can enhance the resilience of food production systems (van Dijk et al., 2021). Such measures include the development of climate-proof crops and the improvement of crop management practices that are better adapted to new climatic conditions and are in line with sustainable policies that reduce negative environmental impacts, such as carbon emissions, and optimize the use of resources (Zhao et al., 2022; Grados et al., 2024). A comprehensive strategy for more sustainable food availability will support economic stability, environmental sustainability, and social equity (Hebinck et al., 2021). Given the increasingly severe impacts of

climate change, the transition to climate-smart agriculture using innovative technologies and modern practices is inevitable and likely one of the only viable paths to future food security.

Developing climate-resilient crops to withstand unpredictable climatic events is a fundamental aspect of climate-adapted agriculture (González Guzmán et al., 2022; Rivero et al., 2022). Plant growth relies on stem cells in meristems, which develop into essential tissues and organs, including leaves, flowers, and roots (Aichinger et al., 2012). These stem cells enable long-term growth and adaptation to environmental changes, contributing to plant resilience against stresses (Lee, 2018). Many differentiated cells can dedifferentiate back into a stem cell-like state when needed. In crops, meristems generate key structures, including fruits and roots, which are vital for agriculture. Research on meristem regulation has greatly impacted agricultural practices, leading to crop varieties that are better adapted to different environments and exhibit enhanced productivity (Lindsay et al., 2024). In addition, differentiated tissues and organs can be protected against unfavorable environmental changes through specific signaling cascades that regulate avoidance and/or tolerance mechanisms that impact, for example, molecular chaperones, photosynthesis, cellular metabolism and gene expression (He et al., 2018; Zhang et al., 2023a) at early and advanced developmental stages (germination, seedling establishment, flowering induction). One should keep in mind that different species, varieties, growth stages and organs display different optimal temperatures and different temperature response and tolerance thresholds (Zhu et al., 2021).

General plant fitness is a major determinant for successful plant reproduction leading to high quality crops. The products of plant sexual reproduction, seeds and fruits, are vital sources of nutrition and energy for a wide range of species including humans. Reproduction is a complex developmental process involving a finely orchestrated sequence of relatively short events and a disturbance at any stage of this chain can seriously hamper reproductive success. Thus, the reproductive process itself and the reproductive organs derived from it are particularly sensitive to stress conditions and failure to successfully complete a reproduction cycle may lead to lower quality and quantity of fruit/seed crops (Chaturvedi et al., 2021). Flowering time is also sensitive to environmental conditions and can be accelerated or delayed by fluctuations in temperature, light quality, photoperiod, and water availability (Antoniu-Kourounioti et al., 2021; Rivelli et al., 2024). For pollinating insect-flower interactions, alterations in flowering time can disrupt the synchronization between the time window for pollination and the availability of pollinators (Čereković et al., 2013; Scaven and Rafferty, 2013; Forrest, 2015). Extreme temperature and precipitation patterns can further affect the timing of fruiting, increase the fruit and seed vulnerability to pathogen attacks, and reduce seed survival and germination (Walck et al., 2011; Liu et al., 2018; Bevacqua et al., 2023).

While global efforts to mitigate the effects of human activities on

climate change are underway, parallel efforts to breed climate-resilient crops and adapt agricultural practices to the new and evolving environmental realities are equally essential to ensure food security and ecosystem sustainability. This article outlines the impact of extreme weather events driven by climate change on plant reproduction and explores strategies to enhance reproductive resilience to support food security worldwide. We propose that a new era of modernization in agriculture that integrates state-of-the-art genetics and genomics, advanced agronomic practices, digital farming technologies, and climate-smart strategies are crucial for addressing these challenges effectively. This holistic approach will involve leveraging genetic engineering, precision agriculture, and artificial intelligence (AI) to optimize resource use, enhance crop resilience, and ensure sustainable agricultural production in the face of climate change.

2. The importance of sexual reproduction for yield of seed and fruit crops

Crop plant reproduction involves several key developmental stages: floral transition, flower development, ovule and embryo sac development, pollen development and pollen tube growth as well as fertilization and seed/fruit development (Dresselhaus and Franklin-Tong, 2013; Hafidh et al., 2016; Mendes et al., 2016). Initially, the transition of a plant from a vegetative to a reproductive stage is characterized by the formation of flowers (Raihan et al., 2021). Flowering is controlled by environmental and endogenous cues including light quality, intensity, photoperiod, temperature, and circadian rhythms, through the action of photoreceptors, clock components and autonomous pathway regulators, many of which converge in controlling the expression of the master regulators of flowering initiation, such as the *Arabidopsis thaliana* FLOWERING LOCUS C (FLC), CONSTANS (CO), FLOWERING LOCUS T (FT), and SUPPRESSOR OF OVEREXPRESSION OF CONSTANS1 (SOC1) genes, to name a few (Maple et al., 2024). Flowering control primarily relies on the activation or repression of the afore-mentioned flowering regulators at the epigenetic, transcriptional and post-translational level. Within angiosperm flowers, the “hidden” haploid male and female gametophytes are formed (reviewed by Butel and Köhler, 2024). The embryo sacs (the female gametophytes) differentiate from megaspore mother cells within ovule primordia and at maturity are typically composed of seven cells including the female gametes, egg and central cell. Ovules grow directly from placenta tissue within the ovary of a pistil. In dicots, ovaries usually embed multiple ovules, whereas monocot flowers contain only a single ovule. Pollen grains (male gametophytes) develop from pollen mother cells in the anther of a stamen. Ovules, embryo sacs and embryos are continuously nourished by the mother plant, while pollen grains are nourished in anther loculi through the tapetum only during its early development. The mature pollen grain has a tough exine cell wall to survive unfavorable environmental conditions in a quiescent mature stage and to reach the stigmatic papillary cells in a viable state. Once there, the pollen grain rehydrates and forms a pollen tube that grows through the transmitting tract until it reaches the ovule and penetrates it via the micropyle, attracted by molecules produced by the synergid cells. At this point, both female and male gametophytes finally meet, the pollen tube bursts, the receptive synergid degenerates and the two male gametes, the sperm cells, are released. During the double fertilization process, one sperm cell fuses with the egg cell given rise to the embryo, while the second sperm cell fuses with the di-haploid central cell forming the triploid endosperm that will nourish the embryo during its development in eudicots and during germination in monocot cereals (Dresselhaus and Franklin-Tong, 2013; Hafidh et al., 2016; Mendes et al., 2016).

Several processes in sexual reproduction are critical for determining the final seed yield *per* fruit and *per* plant. Variations in flower structure, such as the gynoecium, including length of style, stigma, replum and the filament of the stamen, can affect pollination efficiency (Wu et al., 2016; Pan et al., 2019b; Lohani et al., 2020; Baranzelli et al., 2024). Pollination

failure can arise from various abiotic factors that result in reduced pollen viability, insufficient pollen amount, unavailability of pollinators, issues with pollen transfer to the stigma, and/or inhibited pollen germination and tube growth. Regardless of the cause, such pollination troubles typically limit reproduction success and may lead to (partial) sterility (Wilcock and Neiland, 2002). For insect-pollinated species, pollination efficiency greatly depends on the abundance of suitable insect species, the development of which can be differently affected by environmental conditions, especially extreme conditions (Sage et al., 2015). Other factors, including ovule number, fitness of female and male gametophytes, synchronization of tissue development, gametophyte maturity, and pollen tube growth through the pistil, are essential pre-fertilization features required to ensure successful and efficient seed set (Dresselhaus and Franklin-Tong, 2013; Cucinotta et al., 2020). During seed development, events affecting endosperm development and embryo morphogenesis can lead to seed abortion, while issues affecting seed maturation and dormancy can increase the risk of premature seed sprouting and seed germination (Young et al., 2004; Chen et al., 2021; Zhu et al., 2021; Mácová et al., 2022). Therefore, these reproductive failures have a direct impact on seed harvest (yield). In fruit crops, the number of seeds may affect fruit size (Bertin et al., 1998; Aleemullah et al., 2000) and shape, as in multilocular fruits such as tomato (Monselise, 2018).

Abiotic stresses can impair fruit setting, growth and ripening, often leading to reduced yield and compromised quality. Irrigation with water that has increased electrical conductivity can cause salinity stress and negatively impact water transport within the plant, limit gas exchange, and reduce fruit size due to inhibited xylem and phloem flow (Li et al., 2023a). Mild heat stress can accelerate ripening whereas severe heat stress can delay ripening (Biggs and Handa, 1988; Miyazaki et al., 2013; Gao et al., 2017; Nakano et al., 2021; Sharma et al., 2023a). In some cases, abiotic stress can be beneficial in terms of fruit quality. In tomato, moderate salinity or drought stress can enhance fruit quality due to the increase of soluble solids, thereby improving fruit taste, even though smaller fruits are produced. Meanwhile, physiological disorders such as blossom end rot might be more frequent under moderate stress (Petersen et al., 1998).

Climate change can significantly affect perennial fruit crops, particularly during flower initiation, leading to a reduced number of flowers in the following year or flower abortion (Čereković et al., 2014). Higher temperatures and drought stress may inhibit floral transition by accelerating vegetative growth and reducing flower production (Čereković et al., 2013; Bogdziewicz et al., 2020). Contrastingly, for crops that require vernalization for flowering, winter warming can delay the onset of flowering through devernization, and cause yield reduction as shown in winter wheat (Dixon et al., 2019; Penfield et al., 2021).

3. Sexual reproduction of crops in the face of climate change

The impact of high temperatures on crop productivity – Elevated ambient temperatures trigger a developmental program known as thermomorphogenesis that includes the elongation of organs and tissues, hyponastic growth, stomatal development, and accelerated flowering (Zioutopoulou et al., 2021; Delker et al., 2022). While acceleration of flowering can help plants to avoid an upcoming hot period, it can also have trade-offs as it shortens the vegetative growth period, potentially leading to smaller plants with less biomass and lower yield potential and may disrupt synchronization with pollinators, affecting pollen and fruit set as well as their quality (Begcy et al., 2019; Jagadish, 2020; Faust and Iler, 2022). Similarly, higher temperatures can speed up photosynthesis rate, and result in overall increase in plant biomass, but also drive modifications in sink-source relations and reduced partitioning in reproductive organs, concomitantly resulting in a lower yield (Suwa et al., 2010; Lesjak and Calderini, 2017).

Beyond a certain temperature threshold, which may vary between species and genotypes, plants experience heat stress, which is typically

manifested by reduced growth, but also affects all stages of plant development and, above all, reproductive development with major consequences on crop yield (Hedhly, 2011; Cucinotta et al., 2020; Gonzalo et al., 2020). During the reproductive phase, high temperatures cause both morphological and functional alterations in male and female reproductive structures contributing to reduced fertility (Fig. 1; Pressman et al., 2002; Hedhly, 2011; Begcy et al., 2019; Jagadish, 2020; Lohani et al., 2020; Miller et al., 2021).

Heat stress impacts pollen quantity and morphology, the architecture of cell wall and, of utmost importance, pollen metabolism (Zinn et al., 2010; Hedhly, 2011; Begcy et al., 2019; Chaturvedi et al., 2021). Developing pollen and growing pollen tubes are probably the most heat stress-sensitive plant organs (Rieu et al., 2017; Gong et al., 2024). In maize, a 2-day heat stress during different stages of pollen development and pollen tube growth can lead to severe reduction in fertility and grain yield (Gong et al., 2024; Lv et al., 2024). Interestingly, heat stress response mechanisms are activated during pollen development under physiological conditions, as a pro-active thermotolerance mechanism in case of an upcoming heat stress incident (Giorno et al., 2010; Fragkostefanakis et al., 2016; Sze et al., 2024). However, limited male thermotolerance capacity does not explain all variation in reproductive outcomes, suggesting that the female part is also significantly affected by heat (Wang et al., 2021a). In wheat, damages in female organs accounted for 34% of the gross fertility loss in plants exposed to a combination of heat and drought stress (Fabián et al., 2019).

The relative importance of female vs male effects for overall fertility is genotype-dependent for a particular crop and is dependent on the severity of the stress. A recent analysis of processing tomato cultivars concluded that the differences in heat tolerance for yield were not linked to pollen viability or pollination efficiency, but to female fertility (Miller et al., 2021). A well-known female heat stress effect for many crops such as tomato and rice is “stigma exertion”, where the stigma of the flower extends beyond the anthers. In tomato stigma exertion impairs self-pollination and fertility, leading to fruit set reduction (Pan et al., 2019a; Wu et al., 2019, 2024; Shang et al., 2021). In rice, however, stigma exertion can have both advantages and disadvantages under high temperatures. In particular, it can enhance pollen receptivity and mitigate heat-induced pollination issues, particularly under high nighttime temperatures, but it can also negatively affect spikelet fertility during high daytime temperatures due to stamen-pistil asynchrony (Qi and Wu, 2022).

Heat stress also reduces stigma receptivity causing other abnormalities such as reduction in nutrient supply from the style to the growing pollen tube and in the production of pollen tube attractants (Wang et al., 2021a). Moreover, heat stress is often accompanied by an increase in the levels of reactive oxygen species (ROS) to supra-optimal levels leading to cellular damage and thereby affecting ovule and pollen development, pollen-stigma interactions, and pollen tube growth and fertilization (Ali and Muday, 2024). Thus, balancing ROS levels by the use of scavengers holds a great potential for mitigating ROS-induced sterility during reproduction (Gong et al., 2024).

Female reproductive traits play an important, though highly understudied role in reproductive heat tolerance, exhibiting variation in both pre-pollination (i.e. the proper development of female structures) and post-pollination (i.e., proper functioning of female structures) traits. Specifically, heat stress leads to morphological abnormalities of female organs, decreased ovule number and increased ovule abortion in various species (Zinn et al., 2010; Jiang et al., 2019).

The effect of low temperatures on crop productivity – Although global warming is expected to increase the length of the growing season and allow cultivation of warmer climate crops beyond their current range of cultivation, there is a higher risk of acute low temperature stress due to unpredictable cold episodes and late spring or early autumn frosts. Cold stress (i.e., temperatures below the growth limit) can cause chilling injury (typically between 0 °C and 12–15 °C) and freezing injury (below 0 °C) (Kidokoro et al., 2022; Soualiou et al., 2022). The effects of

chilling stress on plants include disruptions to cell membrane fluidity (Thomashow, 1999). Additionally, chilling stress reduces the activity of enzymes such as RuBisCO, leading to decreased CO₂ fixation and reduced energy production (Rivero et al., 2022). This results in stunted growth and delayed development, especially impacting germination and early seedling growth. Chilling temperatures can also disrupt the production and action of growth hormones like gibberellins and auxins, altering growth patterns and developmental processes (Kidokoro et al., 2022). In contrast, freezing stress results in ice formation within plant tissues, thereby causing more severe damage than chilling stress (Sanghera et al., 2011). Ice crystals can physically damage cellular structures, leading to cell death. As ice forms, water is extracted from cells, leading to dehydration and concentration of toxic solutes in the cell sap (Ide et al., 1998). Freezing temperatures halt metabolic processes such as photosynthesis, respiration, and enzymatic functions. Structural damage extends beyond the cellular level, impairing the vascular tissues of plants which affects nutrient and water transport, potentially leading to the death of the plant if exposed to low temperatures for prolonged periods (Ide et al., 1998).

Freezing temperatures disrupt the development of crops, affect the timing of bud break and flowering in fruit trees, leading to a desynchronization between the flowering period and the availability of active pollinators (Thakur et al., 2010), subsequently causing reduced fruit yield (Fig. 1). Additionally, crops that experience early or late season frosts may suffer from reduced vigor and delayed growth cycles, affecting overall productivity (Sanghera et al., 2011). Rice is sensitive to low temperatures and when cold stress occurs during the reproductive phase, it causes significant damage to grain production (Thakur et al., 2010). In wheat, a temperature drop of just 1 °C below the threshold during reproduction can result in production losses ranging from 10% to 90% (Ji et al., 2017), while in tomato cold stress induces malformed fruits (Wu et al., 2023).

The combination of chilling stress with high light conditions disrupts thylakoid electron transport, reduces RuBisCO activity, and induces stomatal closure, thereby decreasing CO₂ uptake (Huang et al., 2023a), and further leads to a reduction in mesophyll conductance (Fu et al., 2016). Beyond a temperature threshold, the reduction in photosynthetic rate becomes irreversible and leads to a decreased source activity, ultimately limiting parental resources for reproduction (Snider et al., 2009; Soualiou et al., 2022). Low temperatures or chilling stress trigger abscisic acid (ABA) signalling, which is essential for cold tolerance due to its capacity to induce stomatal closure and osmolyte biosynthesis (Thomashow, 1999; Raza et al., 2023). Other phytohormones, including ethylene, gibberellins, and auxin, also play roles in cold stress signaling (Lange et al., 2020; Tang et al., 2022; Huang et al., 2023a; Liu et al., 2024). Low temperatures often delay flowering, causing seeds to develop under suboptimal conditions later in the growing season. This leads to high rates of flower abortion, as documented in chickpea (Croser et al., 2003) where reduced ovule size, viability, and missing embryo sacs impair pistil function (Srinivasan et al., 1999). Nevertheless, the female fertility in key warmer climate crops, e.g. rice and maize, exhibits a relatively high cold stress tolerance (Hayase et al., 1969; Dupuis and Dumas, 1990).

Cold stress causes pollen sterility by disrupting sugar metabolism in the tapetum, thereby preventing starch accumulation in pollen grains (Oliver et al., 2005; Huang et al., 2023a). This sterility is signaled by ABA, which down-regulates genes coding for cell wall invertase and monosaccharide transporters (Oliver et al., 2007). Low temperatures also hinder pollen germination and inhibit pollen tube elongation which can lead to severe reduction in fertilization success (Parrotta et al., 2019).

The impact of cold stress extends to grain filling, delaying the cell cycle and reducing cell division rates in species like maize (Rymen et al., 2007). In chickpea, exposure to cold during the pod-filling stage impairs seed filling, increases flower and pod abortion, and significantly reduces seed growth rate, quality, and yield, with late pod-filling stage stress

Effects of abiotic stresses on reproduction**Male gametophyte development**

Disruption of cell division
 Impaired tapetum and carbohydrate accumulation
 Abnormal microspores
 Reduced energy production in anthers
 Reduced pollen production
 Male sterility

Female gametophyte development

Ovule abortion, decreased ovule number.
 Nutrient flow disruption from the style to the ovule
 Incomplete embryo sac development and reduced viability
 Delayed ovule development, malformed/incomplete embryo sacs

Pollination

Reduction/inhibition of pollen tube growth due to ROS production
 Damages to style
 Reduced energy production and metabolism in the style
 Stigma exertion and impaired self-pollination
 Asynchrony between pollen shedding and stigma receptivity
 Asynchronization of flowering with pollinator activity
 Less receptive stigma due to dessication or high moisture
 Reduced flower size and quality, reduced attraction of pollinators
 Sodium accumulation inhibits pollen adherence and pollination efficiency

Seed and embryo development

Reduced seed set due impaired embryo viability
 Smaller seeds, less viable
 Lower germination rate
 Aborted seeds
 Delayed embryo development

Fruit growth and development

Lower fruit set / fruit abortion
 Growth inhibition/reduction
 Lower fruit quality, reduced assimilate accumulation
 Physiological disorders, malformed fruits
 Acceleration/delay of fruit ripening

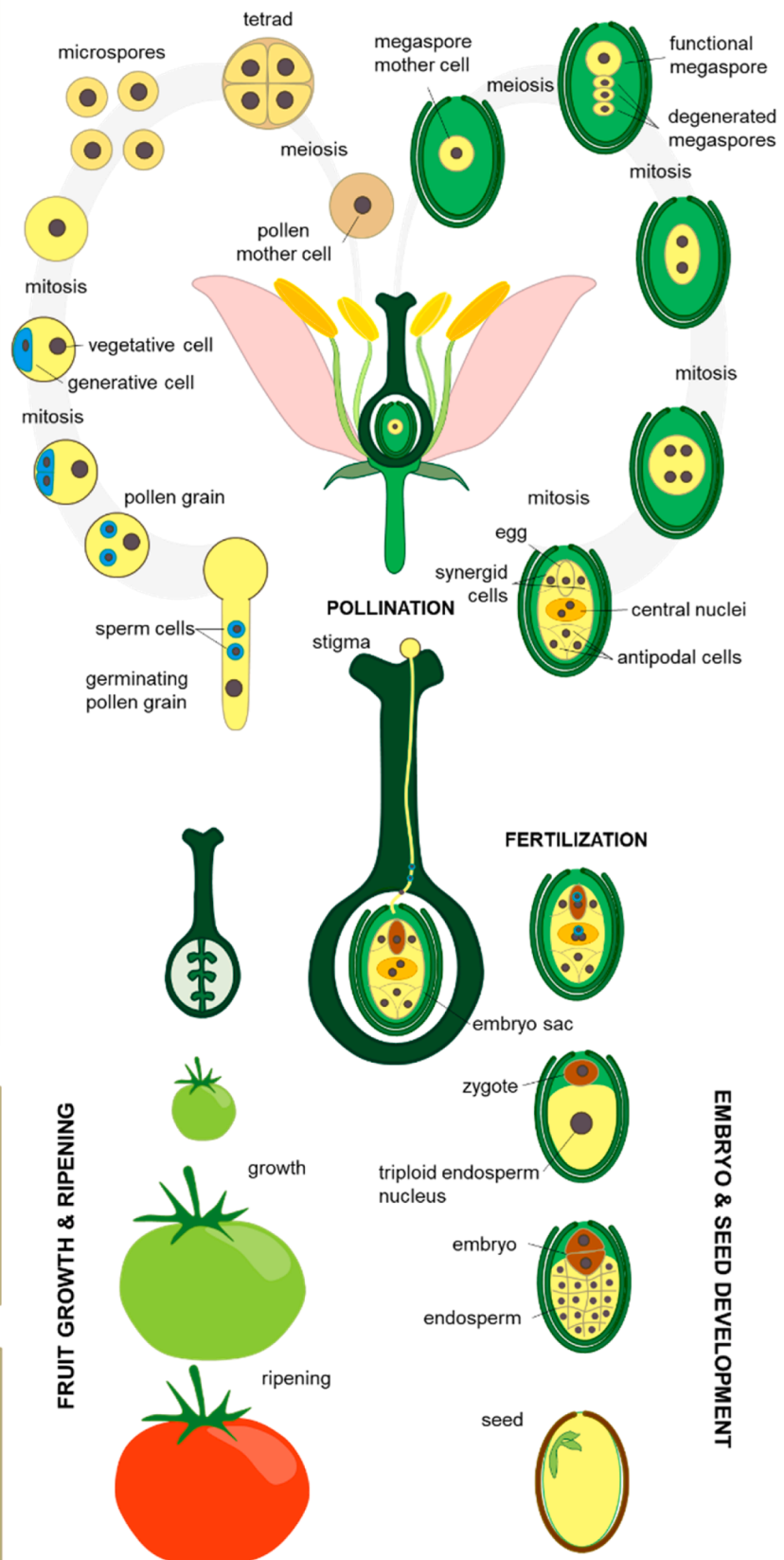
MALE GAMETOPHYTE**FEMALE GAMETOPHYTE**

Fig. 1. Impact of abiotic stresses on plant sexual reproduction. Left: effect of abiotic stresses on different stages of plant reproduction. Right: Reproduction (male and female gametophyte development) including fruit growth and development (tomato as an example) and seed development. Note that pollen grain in species is bicellular (e.g. tomato) and in others tricellular (e.g. grasses). The scheme is exemplary and variations in different species exist that are not shown. Part of the scheme is inspired by Shutterstock image ID 1,024,272,037.

showing a more pronounced negative effect on seed composition and weight compared to early pod-filling stage stress (Kaur et al., 2008). Additionally, chilling temperatures inhibit the accumulation of storage proteins, minerals, and amino acids, reducing both the rate and duration of grain filling (Nayyar et al., 2007). Overall, cold stress limits resource availability for growth, and affects multiple stages of plant reproduction, from flower development to grain filling, leading to significant reductions in crop yield and quality across various economically important crops.

The impact of drought stress on crop productivity – Climate change is associated with the increased frequency and severity of drought periods in different regions of the world (Field et al., 2012). Plant vegetative growth is often severely affected by drought stress but relatively only few studies have reported the effects of drought stress on reproductive organs (Barnabás et al., 2008; Turc and Tardieu, 2018). Drought stress impairs physiological processes such as photosynthesis, and nutrient uptake. In maize, drought stress can cause up to 40% grain yield loss showing that wheat is particularly sensitivity to drought at the flowering and grain-filling stages (Daryanto et al., 2016; Webber et al., 2018; Cerekovic et al., 2024). Additionally, drought often increases the susceptibility of crops to specific pathogens, further contributing to yield loss (Singh et al., 2023).

Drought stress leads to a series of physiological and morphological changes (Su et al., 2013; Čereković et al., 2014; Sallam et al., 2019; Sheoran, 2022; Duvnjak et al., 2024). In drought-stressed plants, sink limitation due to curbed growth, and source limitation resulting from reduced photosynthesis may reduce plant growth and development, thereby adversely affecting crop yield and quality (Rodrigues et al., 2019; Sallam et al., 2019; Hageman and Van Volkenburgh, 2021).

Drought significantly affects the reproductive processes of crops, yet this critical aspect has received far less attention compared to the impact on vegetative tissues. A major effect is the reduction in flower development time, resulting in smaller flowers with lower quality and quantity of nectar, leading to the attraction of fewer pollinators (Fig. 1; Al-Ghzawi et al., 2009). Additionally, drought leads to pollen grain sterility, particularly when stress occurs during meiosis and mitosis, resulting in decreased pollen viability, germination, and pollen tube growth (Al-Ghzawi et al., 2009; Fang et al., 2010). The underlying causes of the damage to these issues include carbohydrate deprivation, increased endogenous ABA concentration, and impaired utilization of sucrose and starch by reproductive sinks (Sehgal et al., 2018). In addition, reduced water availability interferes with normal embryo sac development, leading to malformed embryo sacs and increased ovule abortion (Barnabás et al., 2008).

Grain filling can be severely affected by drought (Farooq et al., 2017). The accumulation of non-reducing sugars to increase the osmotic potential as an acclimation response to drought and failure to build up starch during this period can lead to ovary abortion and reduced grain set and yield (Neumann Andersen et al., 2002). The activity of acid invertase, a crucial enzyme in grain development, is reduced under low water potential, making it a limiting factor in starch biosynthesis and ovary development (Boyer and Westgate, 2004; Farooq et al., 2017). Consequently, inadequate energy sources and elevated ABA levels during drought are key contributors to poor seed set (Liu et al., 2005). Reduced water availability can limit the translocation of assimilates to the developing grains (Španić et al., 2023), resulting in smaller grain size and lower grain mass with overall reductions in crop yield as shown for wheat (Duvnjak et al., 2024).

The impact of flooding and hypoxia stress on crop productivity – The erosive force of climate-induced precipitation is the leading cause of global soil degradation (Borrelli et al., 2023; Panagos et al., 2024). Waterlogging caused by extreme rainfall and flooding creates oxygen-deficient conditions with changes in soil redox status, accumulation of toxic substances that harm plant roots and aerobic soil organisms, affecting nutrient balance and soil properties (Bramley and Tyerman, 2010; Tyagi et al., 2023). The inhibition of root physiological

functions and impaired photosynthesis due to waterlogging stress have adverse effects both on vegetative and reproductive growth, leading to yield loss or even complete harvest failure (Zhou et al., 2020; Pan et al., 2021). Plants cope with stress caused by waterlogging through alterations in morphological traits such as development of distinct root structures and physiological responses including the regulation of hormone levels (Daniel and Hartman, 2024).

The impact of waterlogging on growth is crop-dependent. Few crops, such as rice, exhibit superior tolerance to waterlogged conditions due to developmental and physiological adaptations that allow them to thrive in flooded environments (Xu et al., 2006; Bailey-Serres et al., 2010; Langan et al., 2022). However, other crops, e.g. barley, maize, wheat, bean, rapeseed etc., are particularly vulnerable to waterlogging, with yield losses ranging from 20% to 50%, depending on the duration and timing of the event (Tong et al., 2021; Manik et al., 2022). Moreover, extreme soil moisture increases microbial pathogen distribution and promotes plant vulnerability to diseases, augmenting agricultural losses (Moslemi et al., 2018). Studies on hypoxia have been focused mainly on roots, while the effects of low oxygen on reproduction is less understood. How redox state under water deprivation conditions affects reproduction is important, considering that for example in maize hypoxic conditions naturally arising during growth affect germ cell fate, while oxidizing environment inhibit cell specification and cause ectopic differentiation in deeper tissues (Fig. 1; Kelliher and Walbot, 2012).

The effect of salinity stress on crop productivity – Soil salinization is one of the main abiotic stresses threatening agricultural production, affecting an estimated 800 million hectares of arable land worldwide (Singh, 2022). Soil salinization is exacerbated by climate change factors such as increased evaporation and reduced rainfall in arid and semi-arid regions, which lead to salt accumulation, while poor management practices, including the use of saline irrigation water, inadequate drainage, and overuse of fertilizers, further contribute to salt buildup in agricultural soils (Uri, 2018). Unlike drought or heat episodes, salinity is a chronic stress, and as such, the impact of salinity on plant performance are well predictable. Most of the cultivated species are glycophytes, i.e. they are sensitive to salt, and their average yield is decreased by 50–80% already under moderate salinity (electrical conductivity of soil water of 4–8 dS m⁻¹) (Zörb et al., 2019). Salinity stress, often a result of irrigation practices and sea-level rise, reduces the ability of plants to take up water and essential nutrients, leading to stunted growth and lower yields (Van Zelm et al., 2020).

Salinity affects plants both through increased soil osmotic potential and raising ion concentration; the osmotic effects resemble drought stress, whereas ionic effects alter plant nutrient status and interfere with cellular biochemical reactions (Munns and Tester, 2008). For example, exposure to sodium can delay flowering and trigger ABA-mediated stomatal closure for water conservation (Cai et al., 2017). In addition, sodium impairs photosynthesis by disrupting the proton-motive force and inhibiting CO₂-fixation (Munns et al., 2020). Na⁺ can be sequestered in vacuoles or transported back to the root via the high-affinity K⁺ transporter HKT1, which unloads Na⁺ to parenchymal cells (Hasegawa, 2013). The endodermis and Casparian strip act as barriers to Na⁺ influx and K⁺ efflux, while sodium that enters through non-selective cation channels is expelled by the sodium proton exchanger SOS1/NHX7 (Bassil and Blumwald, 2014). Salinity stress reduces meristem size and primary root elongation by lowering local auxin levels (Liu et al., 2015).

Despite its importance for yield, the impact of salt stress on the development of reproductive organs remains poorly understood. In most crop species, salinity delays flowering, reduces the number of inflorescences and flowers, increases flower abortion and reduces stigma receptivity, pollen production and viability (Fig. 1; Khatun and Flowers, 1995; Khan and Abdullah, 2003; Zhao et al., 2021; Atta et al., 2023).

Primary metabolites, including sugar alcohols, raffinose family oligosaccharides (RFOs; Filippou et al., 2021) and proline, function as osmolytes under salt stress and constitute a core mechanism in the salinity stress response (Flowers et al., 2015; Mattioli et al., 2020).

Plants limit the entry of toxic ions into the reproductive organs compared to other organs (Bigot et al., 2023). In tomato, Na^+ accumulated in style, ovaries and anther intermediate layers but not in the tapetum nor in the pollen grains under salinity stress conditions (Ghanem et al., 2009) and Na^+ concentration was lower in the reproductive organs of the flower than in sepals and petals (Bigot et al., 2023).

4. Genetic approaches to crop improvement

4.1. Utilizing natural genetic variation to improve crop resilience

Climate-proof agriculture requires above all resilient crop varieties. Genetic diversity is a primary resource for breeding programs. It encompasses the genetic variability among lines and genotypes within a species, including different alleles, and allelic interactions (Salgotra and Chauhan, 2023). The focus on maximizing yield and the reduction of genetic diversity in modern domestication practices have led to the development of crops and agricultural systems that have limited adaptive plasticity and are susceptible to climate change (Keneni et al., 2012; Niinemets, 2020; Krug et al., 2023). Indeed, modern cultivars of domesticated cereals are less tolerant to environmental stresses than their wild relatives (Toulotte et al., 2022).

Genetic diversity analysis may be an indicator for breeders to evolve genotypes with diverse genetic components in order to achieve stability in wheat production and get favorable heterotic combinations (Spanic et al., 2016). Therefore, expanding and utilizing the natural genetic variation in breeding programs is crucial for addressing emerging agricultural challenges. Genetic resources, including crop wild relatives (CWR), landraces, and modern varieties, are vital reservoirs of natural genetic variation.

Nowadays, advancements in next-generation sequencing and high-throughput phenotyping enable the effective utilization of plant genetic resources, especially in analyzing complex traits such as abiotic stress tolerance (Weckwerth et al., 2020; Salgotra and Chauhan, 2023; Chaturvedi et al., 2024). A Genome-Wide Association Study (GWAS) on a maize landrace-introgression pool developed from 20 populations collected in arid regions identified two markers associated with grain yield under drought conditions (Barbosa et al., 2021). In cotton, a transcriptome-wide association study across more than 200 accessions identified the protein tyrosine kinase GhHRK1 to be associated with heat stress induced male sterility (Ma et al., 2021). Despite the encouraging results, the explanatory power of genome-wide studies is still very limited, and challenges remain in translating these findings into practical breeding outcomes.

The release of crop genome assemblies has opened up new possibilities for mining alleles that confer tolerance to abiotic stresses. Assemblies are now available for large and complex genomes such as sunflower (3600 Mbp) (Badouin et al., 2017), maize (2500 Mbp) (Schnable et al., 2009), barley (5100 Mbp) (Mayer et al., 2012) (Mayer et al., 2012), and hexaploid bread wheat (17,000 Mbp) (Lukaszewski et al., 2014; Cavalet-Giorsa et al., 2024). Analyzing the pan-genome diversity (representing the complete set of genes within a species) of crop relatives provides valuable insights into gene variation within secondary crop gene pools. For example, about 10% of the cultivated sunflower pan-genome and 1.5% of genes originate from introgressions from wild sunflower species (Hübner et al., 2019). Tomato wild species, such as *Solanum pennellii*, carry stress-tolerant traits, which are crucial for breeding programs and can aid *de novo* domestication efforts, allowing breeders to develop entirely new varieties that are better adapted to environmental challenges (Bolger et al., 2014).

Genebank collections play a crucial role for preserving the genetic diversity that underpins modern crop improvement efforts (Aubry, 2023). These collections house a vast array of genetic resources, including wild relatives, landraces, and traditional cultivars, which are essential for expanding the gene pool available for breeding programs. They serve as the foundation necessary for identifying and incorporating

beneficial traits into modern cultivars (Mascher et al., 2019). The genetic diversity stored in Genebanks ensures that breeders have access to a wide range of alleles, enabling the development of resilient crops that can adapt to the increasing environmental challenges. The Focused Identification of Germplasm Strategy (FIGS) is used to select accessions or genotypes stored in Genebanks that have evolved in environments under natural selection pressure and are therefore likely to carry new allelic variation for useful traits (Haupt and Schmid, 2020). A combination of genomics and information on the evolutionary relationship between the trait and the environment with FIGS has the potential to improve the efficiency and accuracy of identifying the best for use in pre-breeding for target traits.

4.2. Breeding strategies for enhanced crop resilience to abiotic stresses

Breeding strategies aimed at enhancing crop resilience to abiotic stresses have become a critical focus in agricultural research. The first step of these strategies is the identification of reproductive traits associated with stress tolerance which is a long and complex process, requiring extensive research and careful selection to ensure the development of robust and adaptable crop varieties. For fruit and seed crops, reproductive traits include flowering, fertility rate, as well as pollen function, fruit and seed set under stress conditions. Traits such as pollen and egg viability, stigma receptivity, and seed filling are pivotal in maintaining productivity under stress. Furthermore, vegetative organs, such as roots, stems, and leaves, also contribute significantly to reproductive stress resilience, and therefore play a crucial role in overall plant health and yield stability. Deep and extensive root systems enhance water and nutrient uptake, and are therefore critical traits for water- and nutrient-limited conditions (Ghatak et al., 2022). Leaves with reduced stomatal density or improved water-use efficiency maintain turgor and photosynthetic activity during stress periods (Bertolino et al., 2019; Katanic et al., 2021). Moreover, stems with enhanced structural integrity and efficient vascular systems support sustained reproductive development by ensuring the transport of nutrients and water, as well as the necessary mechanical strength to support leaves and reproductive organs (Agustí and Blázquez, 2020).

The evolution of genetic techniques from traditional to modern breeding methods has revolutionized the development of stress-resilient crops. Traditional breeding, relying on phenotypic selection and cross-breeding, laid the foundation for initial improvements in stress tolerance. However, the development of molecular genetics, genomics, and biotechnological tools has accelerated the identification and incorporation of stress-resilient traits, speeding up the development of varieties with enhanced resilience to abiotic stresses. Even more, the identification of signature profiles that are associated with thermotolerance such as the phosphorylation state of a specific set of proteins could aid in the identification of targets for breeding or targets for gene editing that will support crop improvement (Fonseca de Lima et al., 2024).

A variety of molecular markers, from Restriction Fragment Length Polymorphisms (RFLP) to Simple Sequence Repeats (SSRs), and Single Nucleotide Polymorphisms (SNPs), have been extensively used to characterize genetic diversity within and between plant populations. These markers have been instrumental in developing core collections in various crops, including rice (Yu et al., 2011), soybean (Kumar et al., 2022), and tomato (Blanca et al., 2012). The advantages of molecular markers make them indispensable for studying genetic diversity and improving crop productivity and resilience. Recently, Pessoa et al. (2023) conducted a study on drought-tolerant *Solanum pennellii* introgression lines screened at the germination stage to investigate whether this tolerance persisted through the vegetative and reproductive stages. Of the five early-stage drought-tolerant lines two maintained their tolerance in later developmental stages. The authors also identified ten candidate genes for drought-tolerance with associated markers available for marker-assisted selection after validation.

4.2.1. Traditional breeding for abiotic stress tolerance

Crop breeders aim to improve abiotic stress tolerance in breeding germplasm and ultimately in commercial varieties and hybrids while maintaining high quality and yield traits. Screening and propagation of the genetic material in traditional breeding takes place at the production site where the resulting crop is located. This means that the development of lines and hybrids takes place in environmental conditions which are similar to those in which the crop is usually grown. The selection of the best performing lines includes higher yields under different environmental conditions that vary from year to year. Under stressful environmental conditions, there is unconscious selection for resilience because it leads to higher yield, whereas in other years with less severe stress conditions, selection for yield may be at the expense of resilience. On this ground, selection in the field is particularly problematic when resilience has a negative effect on yield (Sadok et al., 2021). The likelihood of a crop being exposed to stress, along with the frequency and severity of that stress, determines whether investing in stress tolerance traits is economically viable and beneficial for long-term crop sustainability. There has been little effort to breed for enhanced phenotypic plasticity that could relax the resilience vs yield trade-off; this largely reflects the overall limited phenotypic plasticity in modern crops (Shaar-Moshe et al., 2019).

Despite the complexity of breeding for reproductive resilience, cultivars from hotter areas, such as parts of India, are often more resilient to heat than those bred in areas with temperate climates. Breeding germplasm, including varieties, hybrids, and landraces, are an important reservoir of genetic diversity and a rich source of resilience traits in high-yielding genetic backgrounds (Lazaridi et al., 2024). Quantitative Trait Loci (QTLs) from breeding germplasm are less likely to be linked to negative pleiotropic effects than traits from wild accessions, known as negative linkage (Huang et al., 2023b). This also means that a resilient germplasm is a very valuable source for academic research to better understand the molecular differences in breeding germplasm.

Breeding companies have already started implementing strategies for adaptation to climate change. These practices involve shifting sowing and planting dates to avoid detrimental effects of drought and high temperatures at flowering and seed set or could involve more logistically complex decisions such as moving the sites of propagation of heat-sensitive crops (e.g. leek) to northern regions in response to low seed yields at the original southern production areas (Hampton et al., 2016). While this may seem an elegant solution, it can come at the cost of the beneficial effect of selection under heat. For example, lettuce seed germination is temperature sensitive (Yoong et al., 2016). However, seeds produced at high ambient temperatures germinate better at higher temperatures than seeds harvested from plants grown at lower temperatures, as the mother plant has a significant influence on seed traits (Penfield and MacGregor, 2017).

4.2.2. Modern breeding techniques

Molecular breeders can interrogate genome sequences using high resolution SNP platforms, genotyping by sequencing (GBS), or even by whole genome re-sequencing. QTLs can be transferred from one cultivar or wild accession to a breeding line (introgression) and the progress towards the generation of near isogenic lines (NILs) can be tracked based on genome sequence (Bita et al., 2011). Genetic data can also be utilized to introduce mutations either derived from mutant populations with randomly generated mutations, or induced targeted mutations generated by gene editing. However, although many genes involved in abiotic stress related processes have been identified, not many stress tolerance loci have been successfully introduced into commercial material. A major reason for this is that reproductive resilience is a complex polygenic trait, as has been demonstrated in crops such as rice, maize, rice, Brassicaceae and tomato (Wen et al., 2019; Cappetta et al., 2021; Seetharam et al., 2021; Li et al., 2023b). The ability of breeders to introduce QTLs into breeding lines and follow them through to a commercial product, alongside all other relevant quality and yield traits, is limited.

This is particularly the case in the breeding of vegetable crops such as tomato, which is bred for many different market segments, each requiring a specific set of traits, including plant height, biotic resistance, fruit size, color, sweetness, flavor, shape, and vine architecture. Therefore, introducing more than one or two QTLs for abiotic stress resilience is often more economically feasible for field crops, such as processing tomatoes.

QTLs for increased pollen viability and pollen number under heat stress have been identified in tomato (Xu et al., 2017). A major QTL conferring higher tolerance under continuous mild heat conditions (31 °C day, 25 °C night) was recently shown to act through enhanced cooling of young flower buds and to improve yield in heat wave-affected commercial production environments (Jansen et al., 2024). A comparative study of QTL and RNA-seq in rice identified 45 heat-responsive genes, 24 of which showed specific expression patterns in reproductive tissues (Raza et al., 2020). *OsBIP2* (Binding Protein 2) was identified as a potential candidate conferring thermotolerance of pollen tubes in response to heat stress. An earlier study provided a landmark of a QTL-identified rice gene, *THERMOTOLERANCE 1* (*TT1*), whose overexpression confers thermoprotection without compromising yield (Li et al., 2015). In addition to the proteasomal *TT1* subunit, chromatin remodeling factors, such as the *Brassica rapa* JUMONJI (*JMJ18*) histone 3 lysine 36 di-/trimethyl (H 3K36me2/3) demethylase were shown to enhance thermal resilience without impairing plant parameters when grown under non-stress conditions (Xin et al., 2024).

The complexity of abiotic tolerance in breeding germplasm was also highlighted by a study of heat stress tolerance in maize that identified 22 QTLs for leaf and tassel thermotolerance. However, tolerance QTLs from different cultivars did not co-localize, indicating that multiple, phenotypically differentiable pathways are involved in the development of foliar stress phenotypes that are relevant to the evaluation of yield reduction after heat stress (McNellie et al., 2018).

One of the biggest challenges in crop breeding is the lengthy time required to develop stable, high-performing varieties, which can take several years. This issue can be addressed by speed breeding which aims to drastically shorten the breeding cycle, allowing the production of multiple crop generations within a year (Haroon et al., 2022; Pandey et al., 2022). Speed breeding is achieved typically by extending the photoperiod and providing continuous light to crops which in combination with single-seed descent method allows 4–6 generations of crops like wheat and barley within a year (Riaz et al., 2016; Hickey et al., 2017; Ghosh et al., 2018). Speed breeding can be complemented by pollen-based selection, where stress application during the pollen development stage helps identify and select for stress-tolerant traits. By exposing pollen to biotic or abiotic stresses, breeders can screen for resilient pollen grains that carry beneficial alleles, which can then fertilize female gametes and produce stress-tolerant offspring (Totsky and Lyakh, 2015; Mohapatra et al., 2020; Singh et al., 2020). Another proposed approach to make crop reproduction tolerant to abiotic stress has been to make the fruit set independent from pollination, and thus potentially bypassing the need for reproductive resilience for a successful fruit set (Klap et al., 2017). A recessive knock-out mutation in the tomato *AGL6* gene, for example, which was found to be the causal natural mutation in the facultative parthenocarpic *pat-k* mutant, resulted in high quality seedless fruits (Gupta et al., 2021). However, although this tomato mutant has been described as facultatively parthenocarpic, seed set in this mutant after manual pollination is exceptionally low, making the trait unsuitable for seed production and commercial application. Another solution may be found in apomixis. While this phenotype is usually discussed in the frame of clonal reproduction (Underwood and Mercier, 2022), it may also render seed and fruit production less sensitive to environmental stress.

The combination of modern breeding techniques such as marker-assisted selection (MAS), advanced target genes identification, as well as the usage of CRISPR/Cas technology (see also below) with speed breeding, the development of resilient high-performing crop varieties

can become more precise and faster than ever before (Haroon et al., 2022; Pandey et al., 2022; Cardi et al., 2023).

4.2.3. Epi-breeding for stress resilience

Epigenetic regulation is crucial for plant stress resilience and reproductive development, as it modulates gene expression in response to environmental stimuli through mechanisms like DNA methylation, histone modification, and non-coding RNA activity (Begcy and Dresselhaus, 2018; Varotto et al., 2020). Epigenetic modifications can lead to both, transient and heritable changes that impact stress tolerance and reproductive success. During reproductive development in cereals, epigenetic alterations, such as changes in DNA methylation and histone modifications, regulate the formation of gametophytes, fertilization, and seed development under stress conditions (Begcy and Dresselhaus, 2018; Dalakouras and Vlachostergios, 2021).

Heat stress affects reproductive tissues in plants through various epigenetic mechanisms. In rice, heat stress disrupts endosperm development by altering the expression of *FERTILIZATION-INDEPENDENT ENDOSPERM 1* (*OsFIE1*), a component of the Polycomb Repressive Complex 2 (PRC2), leading to failure in cellularization due to changes in DNA methylation and histone modifications (Folsom et al., 2014). Additionally, during male germline development, heat stress reduces expression of the *MULTIPLE ARCHESPORIAL CELLS 1* (*MAC1*) gene in barley, potentially regulated by chromatin modifications, leading to anther abortion (Abiko et al., 2005; Wang et al., 2012). In submergence-stressed rice, H3K4me3-mediated regulation of *ALCOHOL DEHYDROGENASE 1* (*ADH1*) expression shifts from di- to tri-methylation and increases H3 acetylation, promoting gene activation (Tsuji et al., 2006).

Epigenetic variation can be subject to selection and result in stably inherited novel phenotypes, suggesting a role in rapid adaptive responses, and therefore presents a valuable tool for improving crop resilience (Schmid et al., 2018). The use of epigenetic variation is explored using epigenetic recombinant inbred lines (epiRILs) (Johannes et al., 2009; Reinders et al., 2009). Screening of an *A. thaliana* epiRIL population based on the mutant parent line *METHYLTRANSFERASE 1* (*met1-3* mutant), which has an almost complete erasure of CG methylation, identified delayed germination of several lines under salinity stress that was carried in the F9 generation, suggesting the stable inheritance of the trait (Reinders et al., 2009). In an *A. thaliana* *ddm1*-based epiRIL population (DDM1: chromatin-remodeling gene Decrease in DNA Methylation) heritable epigenetic variation was demonstrated in drought experiments (Johannes et al., 2009; Zhang et al., 2013). In tomato, mutation in one of the two *DDM1* paralogues resulted in increased thermotolerance in response to mild chronic heat stress, while silencing of *MutS HOMOLOG1* (*MSH1*) in tomato positively affected fruit weight and fruit number under high temperature field conditions (Yang et al., 2015; Singh et al., 2021b). By combining epiRILs with epigenetic mapping techniques like epigenetic quantitative trait loci (epiQTL) analysis, key epigenetic loci associated with desirable traits can be identified, paving the way for epi-breeding (Zhang et al., 2013). Therefore, through the selective application of epialleles, epiRILs can revolutionize crop improvement strategies for stress resilience.

4.3. Generation of climate-proof crops using new genomic techniques (NGTs)

Gene editing is a fascinating area of study in plant breeding and molecular biology. Gene editing can precisely modify specific genes to improve desirable traits, such as stress tolerance and crop yield, by introducing targeted mutations. New Genomic Techniques (NGTs) are a set of techniques that have emerged during the last 20 years and are used to alter the genetic material of an organism (Iqbal et al., 2020). NGTs include gene editing methods such as CRISPR-Cas, Transcription Activator-Like Effector Nucleases (TALENs), and Zinc Finger Nucleases (ZFNs) which enable the precise manipulation of genomes, facilitating

the incorporation of beneficial traits that confer stress tolerance in crops (Pixley et al., 2023).

There is a plethora of examples from studies in crops that used NGTs and achieved increased stress resilience. CRISPR/Cas9 technology was used to create novel variants of the maize *ARGOS8* gene, a negative regulator of ethylene responses, by inserting the *GOS2* promoter into the gene's 5'-untranslated region (Shi et al., 2017). These CRISPR-edited variants exhibited elevated *ARGOS8* transcript levels and grain yield increase without any yield loss under well-watered conditions. Similarly, advances in genome editing have shown promising results in improving drought tolerance in other crops such as tomato. Knockout of the gene coding for the transcription factor LATERAL ORGAN BOUNDARIES DOMAIN 40 (*SILBD40*) in tomato increased drought tolerance (Liu et al., 2020). CRISPR/Cas9 mutagenesis of the ABA receptor *OsPYL9* in rice resulted in plants with higher ABA levels, anti-oxidant activities, chlorophyll content, leaf cuticular wax, and lower stomatal conductance and transpiration rates and consequently with increased drought tolerance (Usman et al., 2020). Mutations in soybean *GmHDZ4* resulted in enhanced growth and improved root architecture with significantly greater root length, surface area, and root tips, which could maintain a higher turgor pressure through increased osmolyte accumulation and higher antioxidant enzyme activity, leading to reduced oxidative stress markers such as hydrogen peroxide, superoxide anion, and malondialdehyde (Zhong et al., 2022).

Furthermore, CRISPR/Cas9 technology was used to enhance salinity tolerance in rice by knocking out the *OsRR22* gene, a B-type response regulator transcription factor involved in cytokinin signal transduction and metabolism, resulting in mutant plants with significantly improved salinity tolerance at the seedling stage without altering other agronomic traits (Zhang et al., 2019). Similarly, a combination of CRISPR/Cas9-mediated editing of the *OsRR22* gene and heterosis utilization led to the development of novel salinity-tolerant rice lines, which maintained agronomic performance comparable to wild types under non-stress conditions (Zhang et al., 2019; Sheng et al., 2023). *OsWRKY63* negatively regulates chilling tolerance in rice and CRISPR/Cas9-generated *OsWRKY63*-knockout mutants showed increased chilling tolerance due to increased expression of cold-induced genes and ROS enzyme coding genes pushing beyond the limits of the genetically encoded capacity for tolerance (Zeng et al., 2020). CRISPR-Cas9 gene editing on three genes in rice, namely *OsPIN5b*, *GS3*, and *OsMYB30*, was used to create new mutants with significantly increased cold tolerance and yield (Zeng et al., 2020). The *ospin5b* mutant resulted in increased panicle length, the *gs3* mutant contributed to larger grain size, and the *osmyb30* mutant provided enhanced cold tolerance, while the triple mutant combined the advantageous traits and cold tolerance. These examples demonstrate that CRISPR-based targeted gene editing can simultaneously modify multiple genes, combining different traits, conferring tolerance to various abiotic stresses. Furthermore, by targeting several genes, polygenic traits can be precisely engineered to enhance complex stress responses. In addition, the use of the CRISPR-Cas technology can be expanded to combine tolerance to multiple plant diseases, as CRISPR-edited intracellular nucleotide-binding leucine-rich repeat receptor *SINRC4a* mutants in tomato exhibited enhanced immunity against fungi, bacteria, and pests without penalty for growth or yield (Pizarro et al., 2020).

Thus, CRISPR-Cas-based mutagenesis is clearly one of the key tools to accelerate the development of climate-resilient crops, crucial for keeping pace with the rapidly increasing severity of extreme climate phenomena. Optimized variants of Cas9 and other emerging Cas systems with increased efficiency, higher specificity and recognition of different Protospacer Adjacent Motif (PAM) sequences currently expand the plant genome engineering toolbox and will be on the frontlines of combating the effects of climate change on crops (Capdeville et al., 2023; Cardi et al., 2023).

CRISPR has been extended beyond genome editing to include CRISPR activation (CRISPRa) and interference (CRISPRi) (Carroll and

Giacca, 2023). Catalytically inactive Cas9 (dCas9) is coupled with transcription activators or inhibitors to regulate gene expression at specific promoters, as well as chromatin modulators for epigenome editing. Targeted activation of *AREB1*, a major drought-responsive transcription factor, by utilizing dCas9HAT (dCas9-Histone acetyltransferase) triggered local chromatin rearrangement by acetylating histone H3K27, hence increasing gene expression during drought (Roca Paixão et al., 2019). A CRISPR/dCas9-based targeted demethylation system was used to remove cytosine methylation at the targeted locus of the *FLOWERING WAGENINGEN (FWA)* promoter (Gallego-Bartolomé et al., 2018). Early flowering allows plants to reproduce and set seeds before adverse environmental conditions arise, improving survival and reproductive success. Therefore, CRISPRa/i provide a potent and precise method to crop development that outperforms traditional breeding and transgenic procedures in terms of efficiency and specificity.

5. Agronomic practices to enhance the resilience of crops to extreme climate

Agronomic practices play a pivotal role in ensuring successful crop reproduction by creating optimal conditions for plant growth and development, which are essential for maximizing reproductive success and yield. In addition to the use of improved genetic material, several agronomic strategies need to be optimized in order to improve yields: (i)

soil management; (ii) timely irrigation; (iii) pest and disease management; (iv) weed control; (v) crop rotation; (vi) proper spacing, planting density and dates of sowing; (vii) fertilizer management; (viii) cover/strip cropping; (ix) other agroecological techniques (Fig. 2; Balchin et al., 2016; Bhaskar et al., 2021; Franke et al., 2022; Juventia et al., 2022; Bogale and Bekele, 2023). As agricultural production is based on these principles, their prioritization is important in order to ensure a robust vegetative growth and a successful reproduction that will provide high yield of seeds and fruits. These practices particularly apply to open field farming systems, or to semi-controlled environment farming, while some are also relevant for modern closed or controlled-environment agriculture systems, such as greenhouses and vertical farms, which are not always immune to the effects of climate change.

Proper water management, including the sustainable use of water resources, maintaining appropriate water quality, and implementing a strategic irrigation plan, is essential for maximizing crop yield and ensuring long-term agricultural productivity. By optimizing these factors, farmers can enhance water use efficiency, support plant health, and mitigate the impacts of water scarcity and environmental stressors on crop production. For example, rising temperatures reduce wheat yields in India, and advancing wheat sowing by addressing delays in the preceding rice growing season, particularly through earlier rice nursery establishment and increasing groundwater irrigation access before monsoon rains, may help mitigate temperature impacts (Umashaanker

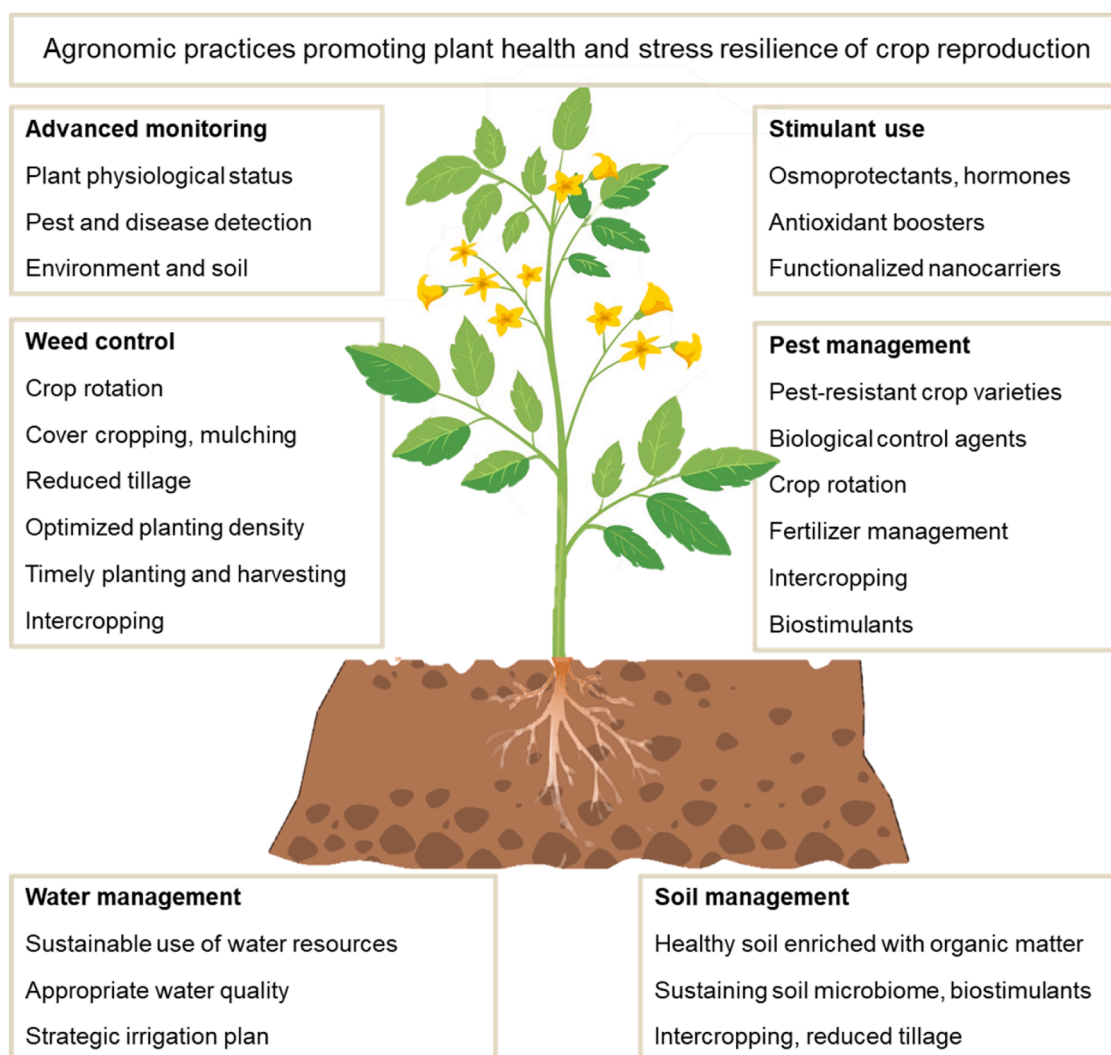


Fig. 2. Overview of agronomic practices that can boost plant health and stress resilience of crop reproduction. Vectors have been designed by Freepik (www.freepik.com) and modified.

et al., 2024).

Maintaining soil health and sustaining the soil microbiome are crucial for boosting crop yields in stressful environments. Healthy soil enriched with organic matter and a diverse microbiome enhances nutrient cycling, improves soil structure, and increases water retention, all of which support plant growth under stress conditions (De Vries et al., 2020). Beneficial soil microbes contribute to enhancing plant resilience in multiple ways, including the formation of self-colonized biofilms around roots, which improve nutrient uptake and protect roots from environmental stresses. They also influence plant physiology by altering root architecture, adjusting stress hormone levels (like ethylene and abscisic acid), and enhancing the production of antioxidants, volatile compounds, and osmolytes (Chatterjee & Niinemets, 2022; Barnawal et al., 2017; Forni et al., 2017; Rafique et al., 2024). A notable example is the root endophytic bacterium *Pseudomonas argentinensis*, strain SA190, which significantly enhances drought tolerance in Arabidopsis and alfalfa by promoting root development and stress response through ABA signaling and epigenetic priming of aquaporin genes (Alwutayd et al., 2023). Thus, utilizing beneficial microbial communities, or targeting the genes and pathways they affect in plants, offers a promising strategy for enhancing crop resilience and productivity in future agricultural practices under stress conditions.

Effective weed management is crucial for enhancing crop stress resilience by reducing competition for resources such as water, nutrients, and light, thereby promoting healthier plant growth and higher yields. Herbicides and tillage are currently the main methods used for weed management, with nowadays well documented negative effects on the environment. In addition, herbicide resistance in crops is an increasing problem for effective weed management. In modern agriculture with an emphasis on sustainable farming practices the adoption of more holistic approaches that encompass practices aimed at reducing weed establishment, growth, and reproduction through methods that enhance the competitiveness of crops and minimize the ecological niches available for weeds (Monteiro and Santos, 2022; Brainard et al., 2023). Key strategies include, among others, crop rotation, cover cropping, reduced tillage, mulching, optimization of planting density and row spacing, timely planting and harvesting, and intercropping which creates a more complex canopy and root structure that suppresses weed growth (Korres et al., 2019). A shift towards biocontrol practices can reduce farmers' dependency on conventional synthetic herbicides, replacing them with botanical herbicides enhanced by modern encapsulation techniques for greater efficacy (Campos et al., 2019).

In the 21st century, eco-friendly farming systems that implement integrated pest management through a better understanding of the underlying causes of diseases and pests, as well as incorporating sustainable agricultural practices, are essential for reducing chemical inputs and enhancing crop resilience (Barzman et al., 2015). This approach promotes the use of biological control agents, crop rotation, and the cultivation of pest-resistant crop varieties, all of which contribute to a healthier ecosystem and more sustainable food production (Stenberg, 2017). Combined with appropriate fertilizer management, intercropping—the practice of growing non-crop plants or alternative crops alongside cash crops—is one of the most straightforward strategies to combat pests and diseases and diversify industrial agriculture. It can be used in several spatial arrangements (Huss et al., 2022). Wide enough for separate cultivation, but thin enough for each crop to interact with other organisms in an ecosystem, is how strip intercropping flourishes. Relay intercropping is the practice of growing many crops or living mulches in succession with some growth cycle overlap (Glaze-Corcoran et al., 2020; Hatt and Döring, 2024).

In addition, incorporating traditional knowledge into modern agronomic practices may be beneficial for enhancing crop resilience and sustainability. Many indigenous farming methods, developed over centuries, emphasize biodiversity, soil health, and water conservation, offering potentially valuable solutions for the current agricultural challenges. Integrating these time-tested practices with modern

techniques may help to create more robust farming systems that are better suited to withstand climate-related stresses. Agroforestry, a practice rooted in traditional knowledge, offers a potentially powerful approach to climate-proof agriculture in some regions such as Africa (Mbow et al., 2014; Pantera et al., 2021). By combining crops, animal husbandry, and trees, agroforestry enhances biodiversity, improves soil health, and increases water retention, all of which contribute to more resilient ecosystems and farming systems (Bogale and Bekele, 2023; Quandt et al., 2023).

By adopting diverse agronomic practices in combination with advanced technologies, farmers can create conditions that enhance the stress resilience of sensitive cultivars but also further boost yields in tolerant varieties. Importantly, these approaches can contribute to sustainability by promoting soil health, reducing the reliance on chemical inputs, and supporting efforts for lower carbon emissions. By optimizing resource utilization and minimizing environmental impacts, these practices help create a more sustainable agricultural landscape that mitigates the effects of climate change and supports long-term food security.

5.1. Optimizing resilience of crop reproduction through chemical applications

Chemical stimulants offer immediate and targeted solutions to a variety of agricultural challenges, including pest control, disease management, and nutrient deficiencies, but can also positively affect crop resilience and productivity under climate challenges (Martínez-Lorente et al., 2024). They are natural or synthetic compounds with the potential ability to enhance plant growth and development under unfavorable environments. Phytohormones such as auxins, cytokinins, ABA, gibberellins, melatonin and jasmonates alter plant growth and productivity under stress conditions (Yakhin et al., 2017). Cytokinins can affect fruit size and yield (Sosnowski et al., 2023) and ethylene releasers such as ethephon can induce flowering, fruit ripening, and leaf abscission (Iqbal et al., 2017). Additionally, synthetic brassinosteroid, glycine betaine, melatonin, polyamines, proline and nitric oxide donors such as sodium nitroprusside can be used to boost yield and improve plant biomass under stress conditions (Ashraf and Foolad, 2007; Simontacchi et al., 2015; Ayub et al., 2021; Hafeez et al., 2021; Shao et al., 2022; Sati et al., 2023; Ullah et al., 2024). Synthetic priming agents can also act as combined donors of compounds known for their priming effects. For instance, NOSH-aspirin (NBS-1120), a novel variant of aspirin that releases both nitric oxide and hydrogen sulfide, was originally developed as an anticancer drug. However, it has also demonstrated potential in mitigating abiotic stress in plants by leveraging the synergistic effects of its active components (Antonioni et al., 2020; Gohari et al., 2024). Similarly, the exogenous application of acetic acid induced the *de novo* jasmonic acid (JA) synthesis and histone H4 acetylation leading to increased tolerance to drought stress (Kim et al., 2017).

Biostimulants on the other hand, such as beneficial microorganisms including mycorrhizal fungi and plant growth-promoting rhizobacteria, humic and fulvic acids, seaweed extracts, amino acids, chitosan, extracts derived from higher plants, or even plant metabolites offer a more sustainable approach (Canellas et al., 2015; Hindangmayum et al., 2019; Priya et al., 2019a; Deolu-Ajayi et al., 2022; Hnini and Aurag, 2024; Martínez-Lorente et al., 2024). They enhance plant growth and development by improving the absorption of nutrients and water by the roots, which is essential for achieving high biomass production and successful harvests (Gilroy and Jones, 2000), stimulating beneficial microbial activity in the soil and enhancing stress tolerance (du Jardin, 2015; Shiade et al., 2024), partly through the regulation of ROS metabolism (Gedeon et al., 2022; Gong et al., 2024). Also, they exert effects at the whole-plant level, influencing physiological processes such as germination, root and shoot growth and morphology, flowering, and fruit set. From the agronomic standpoint, these changes are highly important, as they are closely linked to crop yield and quality, thus determining economic

profitability (Geelen and Xu, 2020). In this way, the use of exogenous biostimulants offers a simple and cost-effective strategy for short-term relief from abiotic stresses and can have immediate positive effects on stress sensitive varieties.

The exogenous addition of salicylic acid (SA) improves pollen mother cell meiosis under heat stress, resulting in increased pollen viability and seed-setting rates in rice (Feng et al., 2018). SA treatment increased freezing tolerance in cold-primed wheat plants (Wang et al., 2020, 2021b). The polyamine spermidine plays a crucial role in pollen germination and stimulates pollen tube growth by promoting protein synthesis under high temperatures (Gupta et al., 2024). Improved pollen viability and germination was also observed in heat stressed tomatoes by exogenous melatonin application (Qi et al., 2018). A brown alga seaweed (*Ascophyllum nodosum*) derived biostimulant significantly enhanced flower development, pollen viability, and overall fruit production in tomato plants under both normal and heat stress conditions (Carmody et al., 2020). This effect was linked to an increased level of simple sugars and the higher expression of heat shock proteins (HSPs), in anthers and young fruits. In tomato plants, treatments with biostimulants coming from higher plants or algae increased fruit production and enhanced the concentration of minerals, bioactive compounds and soluble solids. The treated tomatoes also had greater antioxidant activity and fruit brightness and redness (Colla et al., 2017; Rouphael et al., 2017). Aminolevulinic acid improved yield in bell pepper under high temperatures when applied in its nano-encapsulated form (Hallaji et al., 2024), while proline treatment enhanced pollen fertility and the overall mungbean plant performance under heat stress by increasing proline level and reducing the oxidative stress damage (Priya et al., 2019a,b). In rice spikelets, exogenous application of ABA can reduce the heat-associated pollen sterility by increasing soluble sugars, starch, and non-structural carbohydrates, boosting the synthesis of the essential for

thermotolerance HSPs, and upregulating sugar metabolism and transporter genes (Rezaul et al., 2019). Flavonols, β -carotenoid and their derivatives have also a positive effect on pollen and consequently fertility (Rao et al., 2024). Additionally, melatonin has been found to improve the reproductive success of rice under heat stress by enhancing antioxidant capacity and reducing oxidative damage in pollen and floral tissues (Barman et al., 2024).

Chemical stimulants can be applied as seed coating material or by direct application on plants to ensure long-lasting effects by sustaining a stress tolerant status for a long period. Recent developments in nanotechnology with engineered nanocarriers have allowed the development of smart delivery systems that are eco-friendly and offer increased efficiency and reduced chemical usage (Gohari et al., 2024). For example, melatonin-functionalized chitosan nanoparticles enhance spearmint tolerance to salinity (Gohari et al., 2023), while putrescine-functionalized carbon quantum dots lead to protection of grapevine plants against cadmium stress (Panahirad et al., 2023). Similarly, seed priming with hydrogen sulfide (H_2S) enhanced drought tolerance in sunflower (Fotopoulos et al., 2013; Ocvirk et al., 2021). Taken together, a growing body of evidence suggests that chemical stimulants are practical and immediately available promising solutions for improving plant reproductive success and yield.

6. Artificial intelligence towards a climate-proof agriculture

Crop improvement approaches can be supported by “big data”, high-throughput technologies and AI which can support integrative analyses of genomic, phenotypic, and environmental data (Fig. 3; Yan and Wang, 2023). The rise of omics technologies and next-generation phenotyping tools, like high spectral imaging, push research beyond traditional limits (Sheikh et al., 2024). These technologies enable reliable analysis and

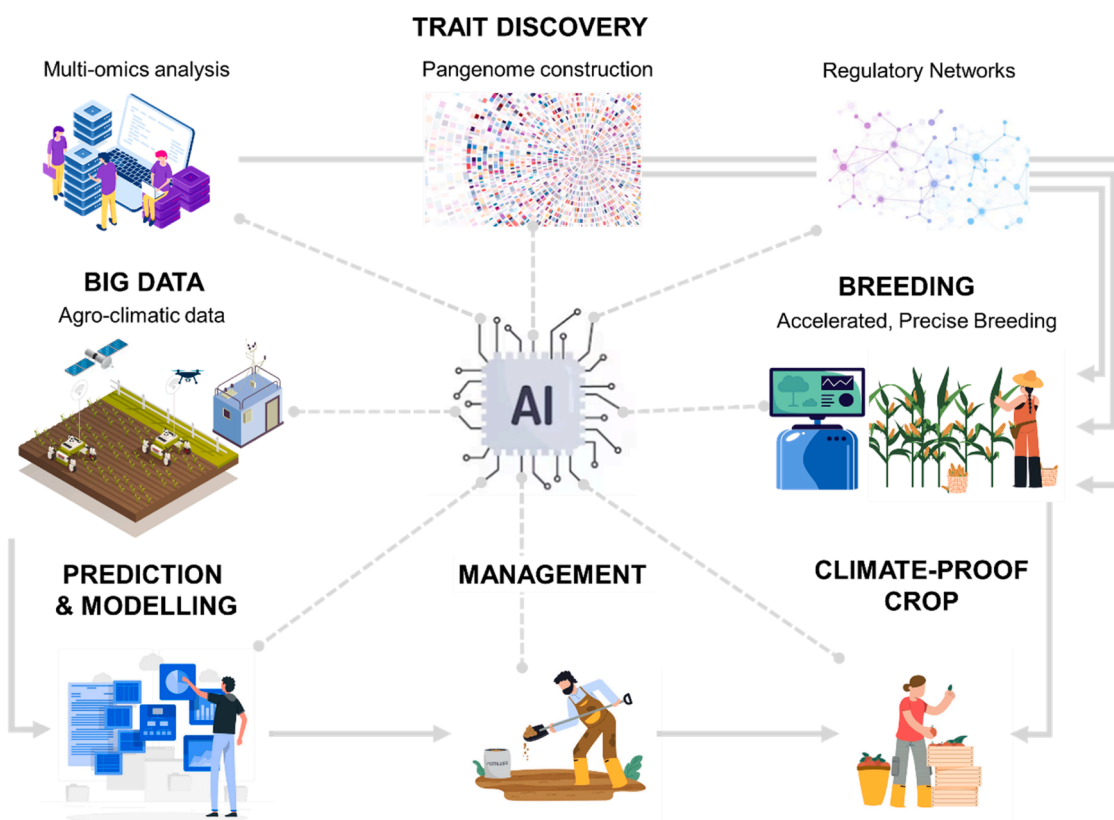


Fig. 3. The role of Artificial Intelligence (AI) towards climate proof agriculture. Solid lines indicate the flow of information and arrows the contribution to specific aspects of science and technology related to agriculture. Dashed lines indicate how AI can collect and process all this information to develop integrated solutions for sustainable, climate-resilient agricultural systems, enabling real-time decision support, optimizing resource management, and advancing stress-tolerant crop varieties. Original vectors have been designed by Freepik (www.freepik.com) and modified.

application of complex data, fostering effective interventions for enhancing crop resilience (Govaichelvan et al., 2024). In the pursuit of transformative and sustainable solutions within almost every aspect of the agriculture sector, AI has emerged as a revolutionary tool as it can perform efficient analysis of vast amounts of data, accurate classification and prediction (Yan and Wang, 2023).

Phenotyping - The traditional methods of phenotyping rely on manual measurements and observations, which are time-consuming, labor-intensive, and often subject to human error. AI-powered image analysis tools, utilizing deep learning algorithms, process large volumes of image data to assess leaf area, plant height, flowering period, and plant's health status, but also flower morphology, pollen viability, and fruit set by analyzing and interpreting images captured by drones or cameras (Walsh et al., 2024). Image-based phenotyping can be combined with Deep Learning (DL) approaches, based on the establishment of artificial neural networks (ANNs) (Singh et al., 2021a). Convolutional neural networks (CNNs) have been mostly applied for static images, while novel approaches based on recurrent neural networks and long-short term memories (LSTMs), have emerged in the last years to provide a dynamic time course of phenotyping studies (Taghavi Namin et al., 2018). Machine Learning (ML) models can be utilized to generate agro-climatic models based on metadata concerning weather, soil, and crop parameters, to predict optimal yield for winter wheat (Mousavi et al., 2023), and to evaluate the effects of flooding on yield based on different vegetation indices (Li et al., 2023c).

CNNs have been used to improve maize tassel detection, ear counting and yield estimation of triticale, soybean yield prediction under different climate and growth conditions, or the determination of plant height in an oilseed rape population, among others (Zou et al., 2020; Alexander et al., 2024; Rodene et al., 2024; Skobalski et al., 2024; Stefański et al., 2024). CNNs are also useful for the evaluation of the performance of reproductive tissues. ML approaches have been recently applied for the image-based discrimination of pollen germination rate and viability and DL has been utilized for the determination of the ideal temperature required for pollen germination (Tan et al., 2022; Yamazaki et al., 2023; Zhang et al., 2023b). Moreover, CNNs have been used to predict the classification of pollen-carrying and pollen-free bees, offering a valuable tool to monitor the effects of climatic change on pollination (Pat-Cetina et al., 2023). Finally, the automatic evaluation of seed germination across crop species has been made possible by DL employing CNNs (Genze et al., 2020; Krzyszton et al., 2024).

-Omics - The integration of ML and AI into plant-omics has revolutionized the analysis and interpretation of complex biological data and helped to overcome barriers like high dimensionality, uncertainty, and non-independence among variables inherent in -omics data (Altman and Krzywinski, 2018). ML and particularly DL can provide more accurate and insightful analyses of plant characteristics affected by genotype and environment interactions (Rai et al., 2019).

In addition, ML is becoming essential in genomics. Evolutionarily informed ML has been used for the annotation of genes and to support the characterization of genomes that are not yet fully understood, aiding breeding studies and genetic approaches (Zia et al., 2023). Given the increasing importance of underutilized crops for food production and the expansion of diversity used for breeding, genomic characterization is crucial. In many cases, developing the pangenome of certain crops is necessary to fully harness their potential for breeding, as it captures the entire genomic diversity within a species, enabling the identification of novel alleles and traits that can be exploited to improve crop resilience and yield. ML and DL enhance the accuracy of variant calling, aiding crop breeding by identifying structural variations and novel alleles (Yocca and Edger, 2022). Haplotype-aware modeling and image-based analyses, enable researchers to extract meaningful patterns from complex data, advancing genomic selection and marker-assisted breeding for crop improvement and even AI-based models can forecast genomic crossovers in maternal and parental plants such as in maize, thereby allowing the identification of genomic regions with elevated mutation

rates (Demirci et al., 2021; Farooq et al., 2024).

Moreover, ML algorithms have been successfully applied to predict gene promoters, cis-regulatory elements, tissue-specific biosynthetic gene expression, and structure and regulation of gene regulatory networks (GRNs), i.e. related to nitrogen signaling (Varala et al., 2018; Uygun et al., 2019). Tools like iDREM use hidden Markov models to reconstruct temporal GRNs in response to biotic and abiotic stresses (Ding et al., 2018; Mishra et al., 2021). The SCENIC analytical pipeline can analyze large datasets efficiently, facilitating cell-specific GRN inference (Van de Sande et al., 2020). A comprehensive maize multi-omics network map was constructed, integrating genomic, transcriptomic, translomic, and proteomic data across development uncovered key regulatory factors for flowering time, including 18 genes with previously unknown roles in this process, and highlighted a histone modification pathway (Han et al., 2023).

The integration of multi-omics data is important because it offers a comprehensive view of biological systems by combining different molecular layers, enabling a deeper understanding of complex gene interactions, improving predictions of phenotypic traits, and facilitating the discovery of biomarkers and regulatory networks for applications like crop breeding and disease research (Picard et al., 2021). Algorithms have been designed and further employed for multi-omics integration, such as multivariate integrative method (MINT) and data integration analysis for biomarker discovery using latent variable approaches for omics studies (DIABLO, based on multiblock sPLS analysis) (Rohart et al., 2017a,b).

Breeding - AI-powered predictive models can simulate how different crop varieties might perform under various climate scenarios, helping researchers anticipate and prepare for future environmental challenges. By integrating data from multiple sources, including field sensors, satellite imagery, and molecular analyses, AI can provide a comprehensive understanding of plant stress responses, guiding more targeted and efficient breeding strategies (Rivero et al., 2022).

AI and ML, such as ML genomic selection, are trained on extensive genomic datasets to identify desirable traits in crops in shorter time frames, than for example traditional MAS, allowing for the acceleration of the breeding process (Tong and Nikoloski, 2021; Chao et al., 2023). Several breeding programs have leveraged these techniques in various crops, including wheat and tomato (Cappetta et al., 2020; Singh et al., 2021c). ML clustering approaches, such as self-organising maps (SOM), can be used for the identification of putative biomarkers as shown for grape (Konecny et al., 2023).

The traditional methods of genomic selection are often parametric, with limited flexibility and cannot sufficiently model the complex non-linear relationship commonly encountered in predicting complex phenotypic traits from genotypes (Sandhu et al., 2021). Thus, a more efficient and assumption-free method that can also capture these non-linear relationships, incorporating different variables and their interaction, in addition to a genomic dataset is necessary to adapt GS analysis to complicated scenarios and hidden patterns in datasets, and reliably predict untested individual progenies from genotypic data (Montesinos-López et al., 2023).

The complexity and scaling issues inherent to omics analysis can be addressed by AI and ML, which facilitate -omics analysis as well as the association of genetic polymorphisms with plant traits (Murmur et al., 2024). Moreover, ML, particularly deep neural networks, can learn nonlinear and complex relationships between genes and environmental conditions, combined with historical data, to make accurate predictions of yields for new hybrids planted in new locations with known weather conditions (Khaki and Wang, 2019). Several agritech companies use AI to develop *in silico* models to identify genetic combinations with the highest likelihood of field success (Waltz, 2017). These models are trained using vast datasets of phenotypic and genomic information. Furthermore, in the era of NGTs, AI and ML can support the development of the best editing strategy by assisting guide RNA selection and target prediction (Alipanahi et al., 2022).

Hybrid breeding is a powerful strategy for crop improvement that enhances genetic diversity, resilience, yield, and trait combinations in response to climate change (Prohens et al., 2017). However, conventional hybrid production through the creation of pure lines can be labor-intensive and challenging. To overcome this, plant breeders utilize double haploid (DH) production to rapidly achieve complete homozygosity in a single generation by inducing androgenesis at the optimal pollen development stage (Thakur et al., 2024). A DL technique called Microscan has been developed to make it easier to select anthers at the optimal developmental stage, hence improving the output of double haploids from anther culture (García-Forkea et al., 2020). DeepTetrad is a DL-based image recognition package that analyzes pollen tetrads for high-throughput measurements of crossover frequency and offer a framework for manipulating the meiotic recombination process in plants to create unique haplotypes with desirable combinations of alleles (Lim et al., 2020). In speed breeding, ML and Neural Networks (NN) are employed to precisely and dynamically control multiple environmental conditions (light, temperature, and humidity) to accelerate plant growth cycles (Sharma et al., 2024). The rate of flowering and seed set can be improved by using ML algorithms and neural networks (Rai, 2022).

Crop management - AI and ML are valuable tools for crop management, as they can perform yield prediction, disease and weed detection, and can assess crop quality (Shaikh et al., 2022). They play a key role in making agricultural production more efficient and stable in the face of global environmental change (Kurek et al., 2007; Niedbala et al., 2024). Contemporary cloud-based systems and Internet-of-Things (IoT) infrastructure, supported by new AI algorithms, facilitate precision agriculture by optimizing the use of resources like water (e.g. the SWAMP project an IoT-based smart water management platform for precision irrigation) (Kamiński et al., 2019), fertilizers, pesticides and herbicides, supporting sustainability and circular economy (Ayoub Shaikh et al., 2022; Shaikh et al., 2022; Sharma et al., 2023b).

By processing data on soil health, remote sensing, weather patterns, and the physiological status of the plant, AI can forecast the best planting times and conditions for optimal reproductive success (Saxena et al., 2022). For instance, AI algorithms have been used to predict the flowering times of various crops by analyzing climatic data and historical growth patterns (Qian-Chuan et al., 2023). Optimizing reproductive resilience to climate change and extreme weather phenomena includes ensuring optimum growth conditions, and AI-driven tools that support systems like Precision Weed Management and Robotics (PWMR) can also be extended to weed control by identifying the most favorable conditions for crops while minimizing the spread and impact of weeds (Monteiro and Santos, 2022). This dual benefit allows for better resource allocation and enhanced crop resilience against both abiotic stressors and weed competition.

7. Integrated approaches as key drivers in the race for enhancing crop resilience

Reaching the goal of increased agricultural production by 28% over the next 10 years to ensure food security requires holistic approaches that utilize advanced genetics, modern agricultural techniques, cutting-edge technology, and sustainable practices (State Food Secur. Nutr. World 2022, 2022; Benitez-Alfonso et al., 2023). This involves exploring plant resilience at the molecular level to identify stress-tolerant traits and breeding new varieties that are better adapted to fluctuating environmental conditions. These strategies are complemented by precision agriculture tools, AI, and data-driven decisions that optimize resource use, enhance crop health, and improve resistance to abiotic and biotic stresses, all of which are essential for meeting future food demands.

Accelerating the breeding process is essential to address the increasing challenges of climate change. The application of a synergized breeding strategy (SBS) including speed breeding, speed vernalization, phenotypic selection, backcrossing and marker-assisted selection has reduced time by around 53% to develop a BC2 near-isogenic line in

wheat (Cha et al., 2024). Better integration of crop genome and pan-genome information along with linkage mapping, association mapping, joint linkage-association mapping strategies and genomic selection, which leads to the identification of the genes and pathways controlling important agronomic traits, can significantly reduce the time for developing new varieties ready for the farm (Graci and Barone, 2023). Bridging the genotype-phenotype gap will allow the identification of marker genes for desired traits. Particularly for resilience of reproduction in different stresses, which is likely a polygenic trait, the integration of multi-omics approaches is necessary together with the analytical power of ML supported by robust statistical models that can assist the integration of these complex datasets, and accelerate trait identification and breeding for resilience in diverse environmental conditions (Weckwerth et al., 2020; Pazhamala et al., 2021; Roychowdhury et al., 2023). Moving from lab to farm is inevitable in order to accelerate breeding, while simulating future climate scenarios with the use of ML and high-throughput phenomic platforms will allow the simultaneous and continuous monitoring of key plant traits as selection markers to identify tolerant genotypes to abiotic stress (Gosa et al., 2019). Such proactive measures can speed up the generation of climate-ready crops.

No doubt, the fastest route to generate stress-resilient crops is through gene editing. De novo domestication through targeted gene editing can fully harness the richness of wild species for improving agronomic traits for crops including stress resilience (Yu et al., 2021). While the commercial use of NGT-based plants is currently restricted in many regions, including the European Union, it is likely inevitable that future generations will reap the benefits of this technology. As regulatory frameworks evolve, the potential of NGTs to revolutionize agriculture will become increasingly recognized and adopted, particularly considering that such crops will require less water, reduced chemical inputs and support better sustainable agriculture and circular economy than many conventional farming methods. This shift will be essential for addressing the growing challenges of food security and environmental sustainability.

It is likely, though, that genetic improvement alone might not be sufficient to meet the growing demands for food security. Therefore, maximizing yields will require the integration of advanced farming practices that optimize resource use, improve soil health, and reduce environmental impact. These practices, such as precision agriculture, optimized irrigation, and sustainable pest management, will work synergistically with stress resilient crops to boost productivity. Combining innovative crop genetics with smart, data-driven agricultural techniques can ensure higher yields, enhance resilience to climate stress, and promote more sustainable food production systems. Introducing sustainable priming practices, such as application of chemical stimulants, introducing diversified cropping systems (crop rotation, cover crops, crops diversification, and intercropping) and/or the use of plant growth-promoting rhizobacteria, arbuscular mycorrhizal and ectomycorrhizal fungi, cyanobacteria, and nitrogen-fixing bacteria will further improve productivity and minimize losses against extreme weather phenomena (De Vries et al., 2020).

In conclusion, addressing the challenges of food security and crop resilience in the face of climate change requires a multifaceted, collaborative approach. It is essential to integrate cutting-edge strategies such as high-throughput phenotyping, genotyping, and gene editing, alongside advanced management practices. This holistic strategy will help us meet the demands of the future, ensuring a more secure and resilient global food system.

The zygote usually divides asymmetrically, forming a large basal cell and a small apical cell. The basal cell develops into the suspensor, a file of cells connecting the developing embryo to the maternal ovule tissues. The apical cell undergoes a series of symmetric and asymmetric cell divisions to form the embryonic shoot-root body axis, including cotyledons, shoot apex, hypocotyl, and root apex (morphogenesis phase).

BOX 1**Double fertilization**

The pollen tube reaches the ovule through the micropyle, penetrates the embryo sac into one of the synergid cells, bursts, and discharges its two sperm nuclei, resulting in the degeneration of the synergid (Hater et al., 2020). Angiosperms exhibit a double fertilization process: one sperm nucleus fertilizes the egg cell, forming a diploid zygote, while the other fuses with the di-haploid central cell forming a triploid endosperm (Baroux and Grossniklaus, 2019). Ovule integuments develop into the seed coat of eudicots, while cereals generate grains or caryopsis representing seeds that additionally contain a pericarp originating from ovary tissue. The coordination of the development of these seed components is vital for a viable seed. The endosperm initially develops as a coenocytic structure, undergoing a series of mitosis without cytokinesis, followed by cellularization. It accumulates starch, oil, and proteins to support embryo development. In eudicots, the endosperm is consumed during seed development, while in monocots, it persists to provide nutrients during seed germination (Doll and Nowack, 2024). The seed coat, a maternal tissue differentiated post-fertilization from the ovule integuments, is influenced by hormonal communication with the endosperm (Figueiredo and Köhler, 2016). Initially, seed coat cells elongate, allowing seed expansion (Zhang et al., 2021). After endosperm cellularization, the seed coat becomes rigid and hydrophobic to protect the embryo from its environment, fixing seed size (Zhang et al., 2024).

During embryo maturation, it accumulates especially storage proteins and lipids, acquiring desiccation tolerance and entering into dormancy (Dresselhaus and Jürgens, 2021; Verma et al., 2022).

After fertilization, the ovary develops into a fruit. In cereals, the pericarp is fused with the thin seed coat to form a caryopsis representing a simple fruit. Fruits protect and nourish developing seeds and facilitate their dispersal. Fruit size often correlates with seed number and is influenced by phytohormones such as auxin, cytokinin and gibberellins (Ruan et al., 2012; Walker and Bennett, 2024). However, parthenocarp, fruit development without fertilization, occurs independently of seed formation. The fruit ripening process can be categorized into climacteric (tomato and peach) and non-climacteric (grapes, strawberries) types. Climacteric fruits rely on a complex interplay of hormones for ripening with ethylene and brassinosteroids playing major roles. In contrast, the onset of ripening in non-climacteric fruits depends strongly on ABA with the intervention of diverse hormonal regulation (Perotti et al., 2023).

This is not one researcher's job but requires interactions from researchers and stakeholders from different disciplines. In a global climate-proof agriculture system, plant biologists and geneticists offer the fundamental knowledge about plant responses; soil scientists and microbiologists provide insights into soil health, nutrient management, and the role of soil microbiome in enhancing plant resilience; and agronomists translate research findings into practical agronomic practices. Biotechnologists develop tools such as biostimulants or customized microbiome mixtures for specific crops and conditions. Data

scientists and AI experts play a central role in this integrated approach, analyzing vast datasets to identify patterns and create predictive models that guide decision-making, while engineers develop new precision agriculture technologies to assist farmers in protecting their crops. Climate experts support this endeavor by assisting in the prediction of climate patterns, which can aid in the development of both short-term and long-term climate adaptation strategies. Further, environmental scientists and ecologists can contribute to the adoption and optimization of sustainable agricultural practices that mitigate environmental impacts.

In a global climate-proof agriculture system economists and policy makers build a local, national and regional framework to support sustainable agricultural practices and ensure they are economically viable for farmers. Social scientists assess the impact of climate change on farming communities, promote farmer education, and facilitate the adoption of resilient farming practices. In this global climate-proof agriculture system, a well-informed and engaged society supports the implementation of these integrated approaches, and fosters innovations that address global challenges with respect to humans, the society and the planet.

BOX 1, BOX 2, BOX 3

CRedit authorship contribution statement

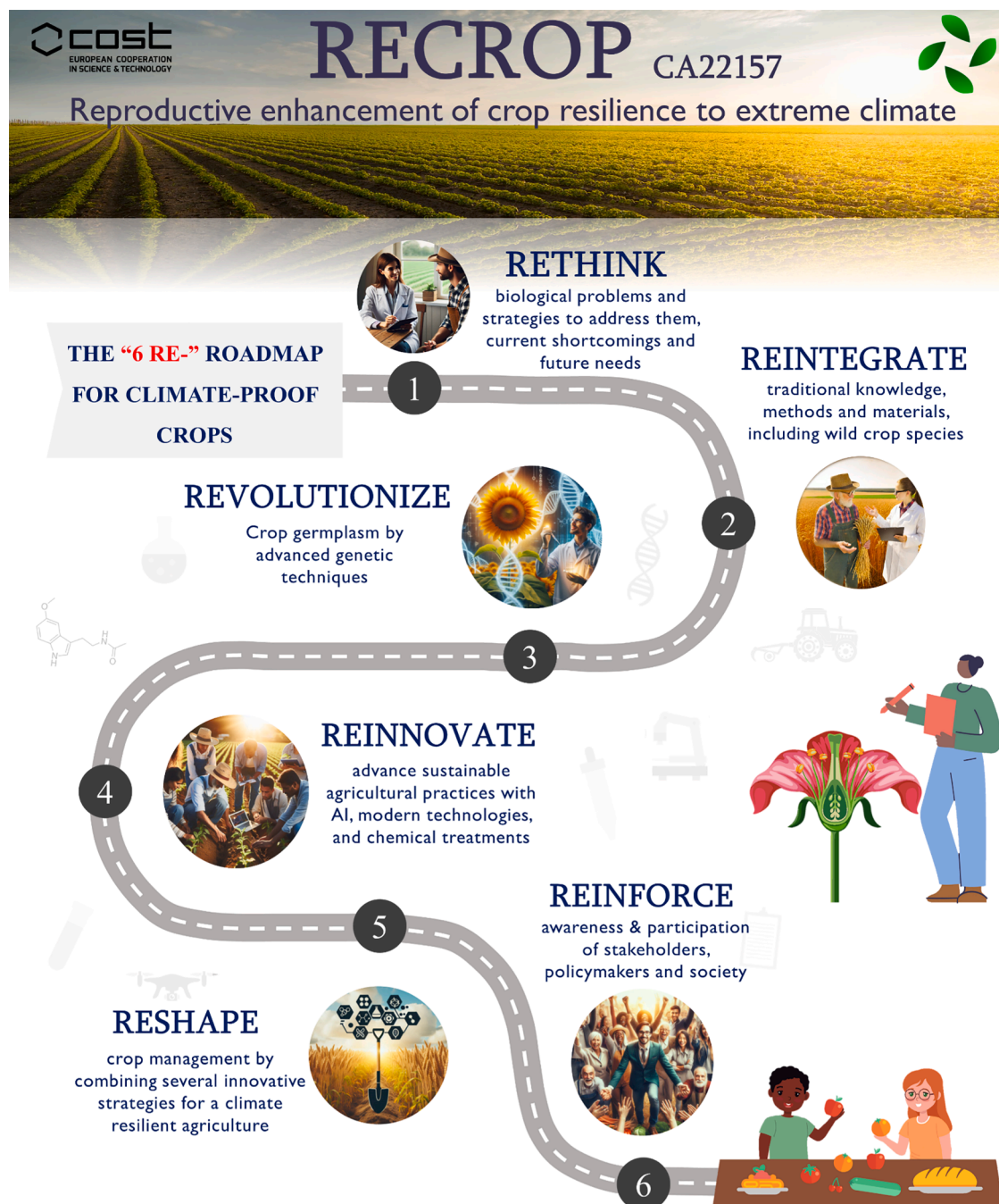
Collins Agho: Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **Adi Avni:** Writing – review & editing,

BOX 2**Integrative Agriculture: Building the climate-proof agriculture of tomorrow**

In the midst of the third decade of the 21st century, advances in science and technology have paved the way for more resilient agricultural practices. The target is clear: ensuring food security while reducing the negative effects of agriculture to climate change. The urgent actions towards this target require the integration of several key research and technology areas in order to enhance crop resilience, optimize resource use, and implement sustainable farming practices. Future integration of multi-omics approaches with high-throughput phenotyping and the assistance from AI and ML can reliably associate genes with desirable traits, while NGTs as well as modern breeding can speed up the development of climate-proof crops of high nutritive value. Of course, the full potential of these new varieties can only be obtained under optimum growth conditions. As discussed in this article, in non-fully controlled environments such as greenhouses, environment stress is often highly unpredictable, and fully optimal conditions are difficult or impossible to achieve. Thus, crop production will benefit from technologies that allow real time monitoring of different environmental and plant characteristics and proper decision-making systems for crop management. Promoting soil health through the use of biostimulants, cover crops, and beneficial microbes improves plant resilience to abiotic stresses and enhances productivity, while the adoption of integrated pest management, precision irrigation, and sustainable fertilization practices reduce the environmental impact of agriculture while maintaining high yields. AI will play an essential role to improve crop management practices, and predict climate impacts on crop yields. Therefore, bringing this knowledge to the farmers is crucial. This can be achieved through extensive education, training programs, and extension services that ensure farmers are equipped with the necessary skills and tools to adopt these advanced practices, ultimately contributing to a more resilient and sustainable agricultural system.

BOX 3

RECROP COST Action: Advancing the Vision of Integrative Agriculture



RECROP is a strategic re-evaluation of agricultural practices to enhance crop resilience and productivity in response to climate change. It involves revisiting and rethinking traditional agricultural methods and incorporating innovative, climate-smart technologies and practices to ensure sustainable crop yields. It includes developing climate-resilient crops by implementing advanced breeding techniques, and adopting agronomic practices that mitigate the adverse effects of climate change. We propose a new era in crop science and technology, where RECROP embodies “six RE-” strategies:

- 1. Rethink biological problems and mitigation strategies:** Critically assess the biological questions underlying the sensitivity of crops to climate change and identify the approaches required to address these problems, and create flexible, long-term strategies that are adaptable to shifting climate patterns and future uncertainties.
- 2. Reintegrate traditional and regional practices:** Combine traditional knowledge with modern innovations to create a more robust agricultural system that can be locally adapted to specific environmental and cultural contexts.
- 3. Revolutionize crop germplasm:** Utilize advanced genetic and genomic techniques to create resilient crop germplasm capable of withstanding extreme weather conditions.

4. **Reinnovate agricultural technologies:** Focus on the continuous development and application of innovative technologies and practical solutions at every level of the agricultural process to address both present and future climate challenges.
5. **Reshape crop management:** Implement new technologies and methods to improve crop management and resilience, adaptable to both smallholder farms and large-scale agricultural systems.
6. **Reinforce societal engagement:** Promote awareness and active participation among communities and policymakers to support innovative genomic solutions, sustainable agricultural practices and climate adaptation strategies. This involves education, policy advocacy, and community-based initiatives to ensure widespread adoption and support for resilient agricultural systems.

RECROP (CA22157; www.recrop-cost.com) is a collaborative initiative supported by the Cooperation in Science and Technology (COST), which aims to advance these strategies by fostering international cooperation, research, and knowledge exchange to build a resilient agricultural future. This initiative brings together scientists, policymakers, farmers, and industry stakeholders to develop and implement innovative solutions that enhance crop resilience and productivity in the face of climate change.

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There is no data presented

References

- Abiko, M., Akibayashi, K., Sakata, T., Kimura, M., Kihara, M., Itoh, K., Asamizu, E., Sato, S., Takahashi, H., Higashitani, A., 2005. High-temperature induction of male sterility during barley (*Hordeum vulgare* L.) anther development is mediated by transcriptional inhibition. *Sex. Plant Reprod.* 18.
- Agustí, J., Blázquez, M.A., 2020. Plant vascular development: mechanisms and environmental regulation. *Cellular and Molecular Life Sciences* 77.
- Aichinger, E., Kornet, N., Friedrich, T., Laux, T., 2012. Plant stem cell niches. *Annu. Rev. Plant Biol.* 63.
- Al-Ghazawi, A.A.M., Zaitoun, S., Gosheh, H., Alqudah, A., 2009. Impacts of drought on pollination of *Trigonella moabitica* (Fabaceae) via bee visitations. *Archives of Agronomy and Soil Science* 55.
- Aleemullah, M., Haigh, A.M., Holford, P., 2000. Anthesis, anther dehiscence, pistil receptivity and fruit development in the Longum group of *Capsicum annum*. *Australian Journal of Experimental Agriculture* 40.
- Alexander, C.S., Yarrow, M., Smith, A., 2024. Who is Responsible For ‘responsible AI’? Navigating challenges to Build Trust in AI Agriculture and Food System Technology. Springer US.
- Ali, M.F., Muday, G.K., 2024. Reactive oxygen species are signaling molecules that modulate plant reproduction. *Plant Cell and Environment* 47.
- Alipanahi, R., Safari, L., Khanteymoori, A., 2022. CRISPR genome editing using computational approaches: A survey. *Front. Bioinform.* 2.
- Altman, N., Krzywinski, M., 2018. The curse(s) of dimensionality this-month. *Nat. Methods* 15.
- Anderson, R., Bayer, P.E., Edwards, D., 2020. Climate change and the need for agricultural adaptation. *Curr. Opin. Plant Biol.* 56.
- Antoniou-Kourounioti, R.L., Zhao, Y., Dean, C., Howard, M., 2021. Feeling Every Bit of Winter – Distributed Temperature Sensitivity in Vernalization. *Front. Plant Sci.* 12.
- Antoniou, C., Xenofontos, R., Chatzimichail, G., Christou, A., Kashfi, K., Fotopoulos, V., 2020. Exploring the potential of nitric oxide and hydrogen sulfide (NOSH)-releasing synthetic compounds as novel priming agents against drought stress in *Medicago sativa* plants. *Biomolecules* 10.
- Ashraf, M., Foolad, M.R., 2007. Roles of glycine betaine and proline in improving plant abiotic stress resistance. *Environ. Exp. Bot.* 59.
- Atta, K., Mondal, S., Gorai, S., Singh, A.P., Kumari, A., Ghosh, T., Roy, A., Hembram, S., Gaikwad, D.J., Mondal, S., et al., 2023. Impacts of salinity stress on crop plants: improving salt tolerance through genetic and molecular dissection. *Front. Plant Sci.* 14.
- Aubry, S., 2023. Genebanking plant genetic resources in the postgenomic era. *Agric. Human. Values.* 40.
- Ayoub Shaikh, T., Rasool, T., Rasheed Lone, F., 2022. Towards leveraging the role of machine learning and artificial intelligence in precision agriculture and smart farming. *Comput. Electron. Agric.* 198.
- Ayub, M.A., Rehman, M.Z.U., Umar, W., Farooqi, Z.U.R., Sarfraz, M., Ahmad, H.R., Ahmad, Z., Aslam, M.Z., 2021. Role of glycine betaine in stress management in plants. *Emerging Plant Growth Regulators in Agriculture. Roles in Stress Tolerance*.
- Badouin, H., Gouzy, J., Grassa, C.J., Murat, F., Staton, S.E., Cottret, L., Lelandais-Brière, C., Owens, G.L., Carrère, S., Mayjonade, B., et al., 2017. The sunflower genome provides insights into oil metabolism, flowering and Asterid evolution. *Nature* 546.
- Bailey-Serres, J., Fukao, T., Ronald, P., Ismail, A., Heuer, S., Mackill, D., 2010. Submergence tolerant rice: SUB1’s journey from landrace to modern cultivar. *Rice* 3.
- Balchin, D., Hayer-Hartl, M., Hartl, F.U., Balch, W.E., Morimoto, R.I., Dillin, A., Kelly, J. W., Gershenson, A., Gierasch, L.M., Pastore, A., et al., 2016. In vivo aspects of protein folding and quality control. *Science* 353, aac4354.
- Baranzelli, M.C., Ochoa-Sánchez, M., Ramos, S.E., Baena-Díaz, F., Sosenski, P., Boege, K., Domínguez, C.A., Fornoni, J., 2024. Effects of anther-stigma position on cross-pollination efficiency in a hermaphroditic plant. *Am. J. Bot.* 111, 1–11.
- Barbosa, P.A.M., Fritsche-Neto, R., Andrade, M.C., Petrolí, C.D., Burguño, J., Galli, G., Willcox, M.C., Sonder, K., Vidal-Martínez, V.A., Sifuentes-Ibarra, E., et al., 2021. Introgression of Maize Diversity for Drought Tolerance: Subtropical Maize Landraces as Source of New Positive Variants. *Front. Plant Sci.* 12.
- Barman, D., Kumar, R., Ghimire, O.P., Ramesh, R., Gupta, S., Nagar, S., Pal, M., Dalal, M., Chinnusamy, V., Arora, A., 2024. Melatonin induces acclimation to heat stress and pollen viability by enhancing antioxidative defense in rice (*Oryza sativa* L.). *Environ. Exp. Bot.* 220.
- Barnabás, B., Jäger, K., Fehér, A., 2008. The effect of drought and heat stress on reproductive processes in cereals. *Plant, Cell and Environment* 31.
- Baroux, C., Grossniklaus, U., 2019. Seeds—An evolutionary innovation underlying reproductive success in flowering plants. *Curr. Top. Dev. Biol.*
- Barzman, M., Barberi, P., Birch, A.N.E., Boonekamp, P., Dachbrodt-Saaydeh, S., Graf, B., Hommel, B., Jensen, J.E., Kiss, J., Kudsk, P., et al., 2015. Eight principles of integrated pest management. *Agron. Sustain. Dev.* 35.
- Bassil, E., Blumwald, E., 2014. The ins and outs of intracellular ion homeostasis: NHX-type cation/H⁺ transporters. *Curr. Opin. Plant Biol.* 22.
- Begcy, K., Dresselhaus, T., 2018. Epigenetic responses to abiotic stresses during reproductive development in cereals. *Plant Reprod.* 31.
- Begcy, K., Nosenko, T., Zhou, L.Z., Fragner, L., Weckwerth, W., Dresselhaus, T., 2019. Male sterility in maize after transient heat stress during the tetrad stage of pollen development. *Plant Physiol.* 181.
- Benitez-Alfonso, Y., Soanes, B.K., Zimba, S., Sinanaj, B., German, L., Sharma, V., Bohra, A., Kolesnikova, A., Dunn, J.A., Martin, A.C., et al., 2023. Enhancing climate change resilience in agricultural crops. *Current Biology* 33.
- Bertin, N., Gary, C., Tchamitchian, M., Vaisiere, B.E., 1998. Influence of cultivar, fruit position and seed content on tomato fruit weight during a crop cycle under low and high competition for assimilates. *Journal of Horticultural Science and Biotechnology* 73.
- Bertolino, L.T., Caine, R.S., Gray, J.E., 2019. Impact of stomatal density and morphology on water-use efficiency in a changing world. *Front. Plant Sci.* 10.
- Bevacqua, D., Vanalli, C., Casagrandi, R., Gatto, M., 2023. A climate-driven compartmental model for fungal diseases in fruit orchards: The impacts of climate change on a brown rot-peach system. *Agric. For. Meteorol.* 332.
- Bhaskar, V., Westbrook, A.S., Bellinder, R.R., DiTommaso, A., 2021. Integrated management of living mulches for weed control: A review. *Weed Technology* 35.
- Biggs, M.S., Handa, A.K., 1988. Biochemical basis of high-temperature inhibition of ethylene biosynthesis in ripening tomato fruits. *Physiol. Plant* 72.
- Bigot, S., Fuksová, M., Martínez, J.P., Lutts, S., Quinet, M., 2023. Sodium and chloride accumulation and repartition differed between the cultivated tomato (*Solanum lycopersicum*) and its wild halophyte relative *Solanum chilense* under salt stress. *Scientia Horticulturae* 321.
- Bitá, C.E., Zenoni, S., Vriezen, W.H., Mariani, C., Pezzotti, M., Gerats, T., 2011. Temperature stress differentially modulates transcription in meiotic anthers of heat-tolerant and heat-sensitive tomato plants. *BMC. Genomics* 12, 384.
- Blanca, J., Cañizares, J., Cordero, L., Pascual, L., Diez, M.J., Nuez, F., 2012. Variation Revealed by SNP Genotyping and Morphology Provides Insight into the Origin of the Tomato. *PLoS. One* 7.
- Bogale, G.A., Bekele, S.E., 2023. Sustainability of Agroforestry Practices and their Resilience to Climate Change Adaptation and Mitigation in Sub-Saharan Africa: A Review. *Ekologia Bratislava* 42.
- Bogdziewicz, M., Kelly, D., Thomas, P.A., Lageard, J.G.A., Hackett-Pain, A., 2020. Climate warming disrupts mast seeding and its fitness benefits in European beech. *Nat. Plants* 6.
- Bolger, A., Scossa, F., Bolger, M.E., Lanz, C., Maumus, F., Tohge, T., Quesneville, H., Alseekh, S., Sørensen, I., Lichtenstein, G., et al., 2014. The genome of the stress-tolerant wild tomato species *Solanum pennellii*. *Nat. Genet.* 46, 1034–1038.
- Borrelli, P., Panagos, P., Alewell, C., Ballabio, C., de Oliveira Fagundes, H., Haregeweyn, N., Lugato, E., Maerker, M., Poesen, J., Vanmaercke, M., et al., 2023. Policy implications of multiple concurrent soil erosion processes in European farmland. *Nat. Sustain.* 6.
- Boyer, J.S., Westgate, M.E., 2004. Grain yields with limited water. *J. Exp. Bot.*
- Bramley, H., Tyerman, S., 2010. Root water transport under waterlogged conditions and the roles of aquaporins. *Waterlogging Signalling and Tolerance in Plants*.
- Butel, N., Köhler, C., 2024. Flowering plant reproduction. *Current Biology* 34, R308–R312.
- Cai, S., Chen, G., Wang, Y., Huang, Y., Marchant, D.B., Wang, Y., Yang, Q., Dai, F., Hills, A., Franks, P.J., et al., 2017. Evolutionary conservation of ABA signaling for stomatal closure. *Plant Physiol.* 174.
- Cammarano, D., Jamshidi, S., Hoogenboom, G., Ruane, A.C., Niyogi, D., Ronga, D., 2022. Processing tomato production is expected to decrease by 2050 due to the projected increase in temperature. *Nat. Food* 3.
- Campos, E.V.R., Proença, P.L.F., Oliveira, J.L., Bakshi, M., Abhilash, P.C., Fraceto, L.F., 2019. Use of botanical insecticides for sustainable agriculture: Future perspectives. *Ecol. Indic.* 105.
- Canellas, L.P., Olivares, F.L., Aguiar, N.O., Jones, D.L., Nebbioso, A., Mazzei, P., Piccolo, A., 2015. Humic and fulvic acids as biostimulants in horticulture. *Scientia Horticulturae* 196.

- Capdeville, N., Schindele, P., Puchta, H., 2023. Getting better all the time — Recent progress in the development of CRISPR/Cas-based tools for plant genome engineering. *Curr. Opin. Biotechnol.* 79.
- Cappetta, E., Andolfo, G., Guadagno, A., Di Matteo, A., Barone, A., Frusciante, L., Ercolano, M.R., 2021. Tomato genomic prediction for good performance under high-temperature and identification of loci involved in thermotolerance response. *Hortic. Res.* 8.
- Cappetta, E., Andolfo, G., Di Matteo, A., Barone, A., Frusciante, L., Ercolano, M.R., 2020. Accelerating tomato breeding by exploiting genomic selection approaches. *Plants* 9.
- Cardi, T., Murovec, J., Bakhsh, A., Boniecka, J., Bruegmann, T., Bull, S.E., Eeckhaut, T., Fladung, M., Galovic, V., Linkiewicz, A., et al., 2023. CRISPR/Cas-mediated plant genome editing: outstanding challenges a decade after implementation. *Trends. Plant Sci.* 28.
- Carmody, N., Goñi, O., Ł., Langowski, O'Connell, S., 2020. Ascorphyllum nodosum Extract Biostimulant Processing and Its Impact on Enhancing Heat Stress Tolerance During Tomato Fruit Set. *Front. Plant Sci.* 11.
- Carroll, M.S., Giacca, M., 2023. CRISPR activation and interference as investigative tools in the cardiovascular system. *International Journal of Biochemistry and Cell Biology* 155.
- Cavalet-Giorsa, E., González-Muñoz, A., Athiyannan, N., Holden, S., Salhi, A., Gardener, C., Quiroz-Chávez, J., Rustamova, S.M., Elkot, A.F., Patpour, M., et al., 2024. Origin and evolution of the bread wheat D genome. *Nature* 633.
- Cerekovic, N., Markovic, M., Radic, V., Cadro, S., Crljenkovic, B., Zapata, N., Paço, T.A., Almeida, W., Stricevic, R., Todorovic, M., 2024. Impact of different water regimes on maize grown at two distinctive pedo-climatic locations in Bosnia and Herzegovina. *Spanish Journal of Agricultural Research* 22, e1201.
- Čerešević, N., Pagter, M., Kristensen, H.L., Brennan, R., Petersen, K.K., 2014. Effects of deficit irrigation during flower initiation of two blackcurrant (*Ribes nigrum* L.) cultivars. *Scientia Horticulturae* 168.
- Čerešević, N., Pagter, M., Kristensen, H.L., Pedersen, H.L., Brennan, R., Petersen, K.K., 2013. Effects of drought stress during flowering of two pot-grown blackcurrant (*Ribes nigrum* L.) cultivars. *Scientia Horticulturae* 162.
- Cha, J.K., Park, H., Kwon, Y., Lee, S.M., Jang, S.G., Kwon, S.W., Lee, J.H., 2024. Synergizing breeding strategies via combining speed breeding, phenotypic selection, and marker-assisted backcrossing for the introgression of Glu-B1i in wheat. *Front. Plant Sci.* 15, 1–9.
- Chao, H., Zhang, S., Hu, Y., Ni, Q., Xin, S., Zhao, L., Ivanisenko, V.A., Orlov, Y.L., Chen, M., 2023. Integrating omics databases for enhanced crop breeding. *J. Integr. Bioinform.* 20.
- Chaturvedi, P., Pierides, I., Zhang, S., Schwarzerova, J., Ghatak, A., Weckwerth, W., 2024. Multiomics for Crop Improvement. In: Pandey, M.K., Bentley, A., Desmae, H., Roorkiwal, M., Varshney, R.K. (Eds.), *Frontier Technologies for Crop Improvement. Sustainability Sciences in Asia and Africa*. Springer, Singapore. https://doi.org/10.1007/978-981-99-4673-0_6.
- Chaturvedi, P., Wiese, A.J., Ghatak, A., Zaveska Drabkova, L., Weckwerth, W., Honys, D., 2021. Heat stress response mechanisms in pollen development. *New Phytologist* 231 (2), 571–585.
- Chen, X., Yoong, F.Y., O'Neill, C.M., Penfield, S., 2021. Temperature during seed maturation controls seed vigour through ABA breakdown in the endosperm and causes a passive effect on DOG1 mRNA levels during entry into quiescence. *New Phytologist* 232.
- Colla, G., Cardarelli, M., Bonini, P., Roupael, Y., 2017. Foliar applications of protein hydrolysate, plant and seaweed extracts increase yield but differentially modulate fruit quality of greenhouse tomato. *HortScience* 52.
- Croser, J.S., Clarke, H.J., Siddique, K.H.M., Khan, T.N., 2003. Low-temperature stress: Implications for chickpea (*Cicer arietinum* L.) improvement. *CRC. Crit. Rev. Plant Sci.* 22.
- Cucinotta, M., Di Marzo, M., Guazzotti, A., de Folter, S., Kater, M.M., Colombo, L., 2020. Gynoecium size and ovule number are interconnected traits that impact seed yield. *J. Exp. Bot.* 71.
- Dalakouras, A., Vlachostergios, D., 2021. Epigenetic approaches to crop breeding: Current status and perspectives. *J. Exp. Bot.* 72.
- Daniel, K., Hartman, S., 2024. How plant roots respond to waterlogging. *J. Exp. Bot.* 75.
- Daryanto, S., Wang, L., Jacinthe, P.A., 2016. Global synthesis of drought effects on maize and wheat production. *PLoS. One* 11.
- Delker, C., Quint, M., Wigge, P.A., 2022. Recent advances in understanding thermomorphogenesis signaling. *Curr. Opin. Plant Biol.* 68, 102231.
- Demirci, M., Gozde, H., Taplamacioglu, M.C., 2021. Comparative Dissolved Gas Analysis with Machine Learning and Traditional Methods. In: *HORA 2021 - 3rd International Congress on Human-Computer Interaction, Optimization and Robotic Applications, Proceedings*.
- Deolu-Ajayi, A.O., van der Meer, I.M., van der Werf, A., Karlova, R., 2022. The power of seaweeds as plant biostimulants to boost crop production under abiotic stress. *Plant Cell and Environment* 45.
- Deryn, D., Conway, D., Ramankutty, N., Price, J., Warren, R., 2014. Global crop yield response to extreme heat stress under multiple climate change futures. *Environmental Research Letters* 9.
- van Dijk, M., Morley, T., Rau, M.L., Saghai, Y., 2021. A meta-analysis of projected global food demand and population at risk of hunger for the period 2010–2050. *Nat. Food* 2.
- Ding, J., Hagood, J.S., Ambalavanan, N., Kaminski, N., Bar-Joseph, Z., 2018. iDREM: Interactive visualization of dynamic regulatory networks. *PLoS. Comput. Biol.* 14.
- Dixon, L.E., Karsai, I., Kiss, T., Adamski, N.M., Liu, Z., Ding, Y., Allard, V., Boden, S.A., Griffiths, S., 2019. VERNALIZATION1 controls developmental responses of winter wheat under high ambient temperatures. *Development (Cambridge)* 146.
- Doll, N.M., Nowack, M.K., 2024. Endosperm cell death: roles and regulation in angiosperms. *J. Exp. Bot.* 75.
- Dresselhaus, T., Franklin-Tong, N., 2013. Male-female crosstalk during pollen germination, tube growth and guidance, and double fertilization. *Mol. Plant* 6.
- Dresselhaus, T., Jürgens, G., 2021. Comparative Embryogenesis in Angiosperms: Activation and Patterning of Embryonic Cell Lineages. *Annu. Rev. Plant Biol.* 72.
- Dupuis, I., Dumas, C., 1990. Influence of temperature stress on in vitro fertilization and heat shock protein synthesis in maize (*Zea mays* L.) reproductive tissues. *Plant Physiol.* 94.
- Duvnjak, J., Katanic, Z., Sarcevic, H., Spanic, V., 2024. Analysis of the Photosynthetic Parameters, Grain Yield, and Quality of Different Winter Wheat Varieties over a Two-Year Period. *Agronomy* 14.
- Fábian, A., Sáfrán, E., Szabó-Eitel, G., Barnabás, B., Jäger, K., 2019. Stigma functionality and fertility are reduced by heat and drought co-stress in wheat. *Front. Plant Sci.* 10.
- Fang, X., Turner, N.C., Yan, G., Li, F., Siddique, K.H.M., 2010. Flower numbers, pod production, pollen viability, and pistil function are reduced and flower and pod abortion increased in chickpea (*Cicer arietinum* L.) under terminal drought. *J. Exp. Bot.* 61.
- Farooq, M.A., Gao, S., Hassan, M.A., Huang, Z., Rasheed, A., Hearne, S., Prasanna, B., Li, X., Li, H., 2024. Artificial intelligence in plant breeding. *Trends in Genetics* xx, 1–18.
- Farooq, M., Gogoi, N., Barthakur, S., Baroowa, B., Bharadwaj, N., Alghamdi, S.S., Siddique, K.H.M., 2017. Drought Stress in Grain Legumes during Reproduction and Grain Filling. *J. Agron. Crop. Sci.* 203.
- Faust, M.N., Iler, A.M., 2022. Pollinator-mediated reproductive consequences of altered co-flowering under climate change conditions depend on abiotic context. *Climate Change Ecology* 3.
- Faye, B., Webber, H., Gaiser, T., Müller, C., Zhang, Y., Stella, T., Latka, C., Reckling, M., Heckelee, T., Helming, K., et al., 2023. Climate change impacts on European arable crop yields: Sensitivity to assumptions about rotations and residue management. *European Journal of Agronomy* 142.
- Feng, B., Zhang, C., Chen, T., Zhang, X., Tao, L., Fu, G., 2018. Salicylic acid reverses pollen abortion of rice caused by heat stress. *BMC. Plant Biol.* 18.
- Figueiredo, D.D., Köhler, C., 2016. Bridging the generation gap: Communication between maternal sporophyte, female gametophyte and fertilization products. *Curr. Opin. Plant Biol.* 29.
- Filippou, P., Zarza, X., Antoniou, C., Obata, T., Villarroel, C.A., Ganopoulos, I., Harokopos, V., Gohari, G., Aidinis, V., Madesis, P., et al., 2021. Systems biology reveals key tissue-specific metabolic and transcriptional signatures involved in the response of Medicago truncatula plant genotypes to salt stress. *Comput. Struct. Biotechnol. J.* 19.
- Flowers, T.J., Munns, R., Colmer, T.D., 2015. Sodium chloride toxicity and the cellular basis of salt tolerance in halophytes. *Ann. Bot.* 115.
- Folsom, J.J., Begcy, K., Hao, X., Wang, D., Walia, H., 2014. Rice fertilization-independent Endosperm1 regulates seed size under heat stress by controlling early endosperm development. *Plant Physiol.* 165.
- Fonseca de Lima, C.F., Zhu, T., Van den Broeck, L., Van De Cotte, B., Locke, A.M., Sozzani, R., De Smet, I., 2024. Large scale comparative wheat phosphoproteome profiling pinpoints temperature-associated molecular signatures. *bioRxiv*. 2024.08.20.608123.
- Forrest, J.R.K., 2015. Plant-pollinator interactions and phenological change: What can we learn about climate impacts from experiments and observations? *Oikos* 124.
- Forster, P.M., Smith, C., Walsh, T., Lamb, W.F., Lamboll, R., Hall, B., Hauser, M., Ribes, A., Rosen, D., Gillett, N.P., et al., 2024. Indicators of Global Climate Change 2023: annual update of key indicators of the state of the climate system and human influence. *Earth. Syst. Sci. Data* 16, 2625–2658.
- Fotopoulos, V., Christou, A., Manganaris, G., 2013. Hydrogen sulfide as a potent regulator of plant responses to abiotic stress factors. *Molecular Approaches in Plant Abiotic Stress*.
- Fragkostefanakis, S., Mesihovic, A., Simm, S., Paupière, M.J., Hu, Y., Paul, P., Mishra, S. K., Tschiersch, B., Theres, K., Bovy, A., et al., 2016. HsfA2 controls the activity of developmentally and stress-regulated heat stress protection mechanisms in tomato male reproductive tissues. *Plant Physiol.* 170, 2461–2477.
- Franke, J.A., Müller, C., Minoli, S., Elliott, J., Folberth, C., Gardner, C., Hank, T., Izaurralde, R.C., Jägermeyr, J., Jones, C.D., et al., 2022. Agricultural breadbaskets shift poleward given adaptive farmer behavior under climate change. *Glob. Chang. Biol.* 28.
- Fu, J., Gates, R.N., Xu, Y., Hu, T., 2016. Diffusion limitations and metabolic factors associated with inhibition and recovery of photosynthesis following cold stress in *Elymus nutans* Griseb. *Journal of Photochemistry and Photobiology B: Biology* 163.
- Gallego-Bartolomé, J., Gardiner, J., Liu, W., Papikian, A., Ghoshal, B., Kuo, H.Y., Zhao, J. M.C., Segal, D.J., Jacobsen, S.E., 2018. Targeted DNA demethylation of the arabidopsis genome using the human TET1 catalytic domain. *Proc. Natl. Acad. Sci. U. S. A.* 115.
- Gao, Y., Wang, Y., Xin, H., Li, S., Liang, Z., 2017. Involvement of Ubiquitin-Conjugating Enzyme (E2 Gene Family) in Ripening Process and Response to Cold and Heat Stress of *Vitis vinifera*. *Sci. Rep.* 7.
- García-Fora, E., García-Pérez, A., Gimeno-Páez, E., Sánchez-Gimeno, A., Vilanova, S., Prohens, J., Pastor-Calle, D., 2020. A deep learning-based system (Microscan) for the identification of pollen development stages and its application to obtaining doubled haploid lines in eggplant. *Biology (Basel)* 9.
- Gedeon, S., Ioannou, A., Balestrini, R., Fotopoulos, V., Antoniou, C., 2022. Application of Biostimulants in Tomato Plants (*Solanum lycopersicum*) to Enhance Plant Growth and Salt Stress Tolerance. *Plants* 11.
- Geelen, D., Xu, L., 2020. The Chemical Biology of Plant Biostimulants.

- Genze, N., Bharti, R., Grieb, M., Schultheiss, S.J., Grimm, D.G., 2020. Accurate machine learning-based germination detection, prediction and quality assessment of three grain crops. *Plant Methods* 16.
- Ghanem, M.E., Van Elteren, J., Albacete, A., Quinet, M., Martínez-Andújar, C., Kinet, J. M., Pérez-Alfocea, F., Lutts, S., 2009. Impact of salinity on early reproductive physiology of tomato (*Solanum lycopersicum*) in relation to a heterogeneous distribution of toxic ions in flower organs. *Functional Plant Biology* 36.
- Ghatak, A., Schindler, F., Bachmann, G., Engelmeier, D., Bajaj, P., Brenner, M., Fragner, L., Varshney, R.K., Subbarao, G.V., Chaturvedi, P., et al., 2022. Root exudation of contrasting drought-stressed pearl millet genotypes conveys varying biological nitrification inhibition (BNI) activity. *Biol. Fertil. Soils* 58.
- Ghosh, S., Watson, A., Gonzalez-Navarro, O.E., Ramirez-Gonzalez, R.H., Yanes, L., Mendoza-Suárez, M., Simmonds, J., Wells, R., Rayner, T., Green, P., et al., 2018. Speed breeding in growth chambers and glasshouses for crop breeding and model plant research. *Nat. Protoc.* 13.
- Gilroy, S., Jones, D.L., 2000. Through form to function: Root hair development and nutrient uptake. *Trends. Plant Sci.* 5.
- Giorno, F., Wolters-Arts, M., Grillo, S., Scharf, K.D., Vriezen, W.H., Mariani, C., 2010. Developmental and heat stress-regulated expression of HsfA2 and small heat shock proteins in tomato anthers. *J. Exp. Bot.* 61, 453–462.
- Glaze-Corcoran, S., Hashemi, M., Sadeghpour, A., Jahanzad, E., Keshavarz Afshar, R., Liu, X., Herbert, S.J., 2020. Understanding intercropping to improve agricultural resiliency and environmental sustainability. *Advances in Agronomy*.
- Gohari, G., Farhadi, H., Panahirad, S., Zareei, E., Labib, P., Jafari, H., Mahdavinia, G., Hassanpouraghdam, M.B., Ioannou, A., Kulak, M., et al., 2023. Mitigation of salinity impact in spearmint plants through the application of engineered chitosan-melatonin nanoparticles. *Int. J. Biol. Macromol.* 224.
- Gohari, G., Jiang, M., Manganaris, G.A., Zhou, J., Fotopoulos, V., 2024. Next generation chemical priming: with a little help from our nanocarrier friends. *Trends. Plant Sci.* 29.
- Gong, W., Oubounyt, M., Baumbach, J., Dresselhaus, T., 2024. Heat-stress-induced ROS in maize silks cause late pollen tube growth arrest and sterility. *iScience* 27, 110081.
- González Guzmán, M., Cellini, F., Fotopoulos, V., Balestrini, R., Arbona, V., 2022. New approaches to improve crop tolerance to biotic and abiotic stresses. *Physiol. Plant* 174.
- Gonzalo, M.J., Li, Y.C., Chen, K.Y., Gil, D., Montoro, T., Nájera, I., Baixauli, C., Granell, A., Monforte, A.J., 2020. Genetic Control of Reproductive Traits in Tomatoes Under High Temperature. *Front. Plant Sci.* 11.
- Gosa, S.C., Lupo, Y., Moshelion, M., 2019. Quantitative and comparative analysis of whole-plant performance for functional physiological traits phenotyping: New tools to support pre-breeding and plant stress physiology studies. *Plant Science* 282.
- Graci, S., Barone, A., 2023. Tomato plant response to heat stress: a focus on candidate genes for yield-related traits. *Front. Plant Sci.* 14.
- Grados, D., Kraus, D., Haas, E., Butterbach-Bahl, K., Olesen, J.E., Abalos, D., 2024. Common agronomic adaptation strategies to climate change may increase soil greenhouse gas emission in Northern Europe. *Agric. For. Meteorol.* 349.
- Gupta, S.K., Barg, R., Arazi, T., 2021. Tomato agamous-like6 parthenocarp is facilitated by ovule integument reprogramming involving the growth regulator KLUH. *Plant Physiol.* 185.
- Gupta, S., Novák, O., Kulkarni, M.G., Doležalova, I., Van Staden, J., Doležal, K., 2024. Unleashing the Potential of Biostimulants in Stimulating Pollen Germination and Tube Growth. Springer US.
- Gupta, R., Somanathan, E., Dey, S., 2017. Global warming and local air pollution have reduced wheat yields in India. *Clim. Change* 140.
- Hafeez, M.B., Zahra, N., Zahra, K., Raza, A., Khan, A., Shaukat, K., Khan, S., 2021. Brassinosteroids: Molecular and physiological responses in plant growth and abiotic stresses. *Plant Stress* 2.
- Hafidh, S., Fila, J., Honys, D., 2016. Male gametophyte development and function in angiosperms: a general concept. *Plant Reprod.* 29.
- Hageman, A., Van Volkenburgh, E., 2021. Sink strength maintenance underlies drought tolerance in common bean. *Plants* 10.
- Hallaji, B., Haghighi, M., Abolghasemi, R., Mozafarian, M., 2024. Effect of foliar applications of aminolevulinic acid (bulk and nano-encapsulated) on bell pepper under heat stress. *Plant Stress* 12, 100477.
- Hampton, J.G., Conner, A.J., Boelt, B., Chastain, T.G., Rolston, P., 2016. Climate change: Seed production and options for adaptation. *Agriculture (Switzerland)* 6.
- Han, L., Zhong, W., Qian, J., Jin, M., Tian, P., Zhu, W., Zhang, H., Sun, Y., Feng, J.W., Liu, X., et al., 2023. A multi-omics integrative network map of maize. *Nat. Genet.* 55.
- Haroon, M., Wang, X., Afzal, R., Zafar, M.M., Idrees, F., Batool, M., Khan, A.S., Imran, M., 2022. Novel Plant Breeding Techniques Shake Hands with Cereals to Increase Production. *Plants* 11.
- Hasegawa, P.M., 2013. Sodium (Na⁺) homeostasis and salt tolerance of plants. *Environ. Exp. Bot.* 92.
- Hater, F., Nakel, T., Gro-Hardt, R., 2020. Reproductive Multitasking: The Female Gametophyte. *Annu. Rev. Plant Biol.* 71.
- Hatt, S., Döring, T.F., 2024. The interplay of intercropping, wildflower strips and weeds in conservation biological control and productivity. *J. Pest. Sci.* (2004).
- Haupt, M., Schmid, K., 2020. Combining focused identification of germplasm and core collection strategies to identify genebank accessions for central European soybean breeding. *Plant Cell and Environment* 43.
- Hayase, H., Satake, T., Nishiyama, I., Ito, N., 1969. Male Sterility Caused by Cooling Treatment at the Meiotic Stage in Rice Plants: II. The most sensitive stage to cooling and the fertilizing ability of pistils. *Japanese Journal of Crop Science* 38.
- He, M., He, C.Q., Ding, N.Z., 2018. Abiotic stresses: General defenses of land plants and chances for engineering multistress tolerance. *Front. Plant Sci.* 871.
- Hebinck, A., Zurek, M., Achterbosch, T., Forkman, B., Kuijsten, A., Kuiper, M., Nørrung, B., 't, Veer P van, Leip, A., 2021. A Sustainability Compass for policy navigation to sustainable food systems. *Glob. Food Sec.* 29.
- Hedhly, A., 2011. Sensitivity of flowering plant gametophytes to temperature fluctuations. *Environ. Exp. Bot.* 74, 9–16.
- Hickey, L.T., Germán, S.E., Pereyra, S.A., Diaz, J.E., Ziem, L.A., Fowler, R.A., Platz, G.J., Frankowiak, J.D., Dieters, M.J., 2017. Speed breeding for multiple disease resistance in barley. *Euphytica* 213.
- Hidangmayum, A., Dwivedi, P., Katiyar, D., Hemantaranjan, A., 2019. Application of chitosan on plant responses with special reference to abiotic stress. *Physiology and Molecular Biology of Plants* 25.
- Hnini, M., Aurag, J., 2024. Genetic diversity, stress tolerance and phytobeneficial potential in rhizobacteria of *Vachellia tortilis* subsp. *raddiana* 3.
- Huang, B., Fan, Y., Cui, L., Li, C., Guo, C., 2023a. Cold Stress Response Mechanisms in *Anther Development*. *Int. J. Mol. Sci.* 24.
- Huang, K., Jahani, M., Gouzy, J., Legendre, A., Carrere, S., Lázaro-Guevara, J.M., González Segovia, E.G., Todesco, M., Mayjonade, B., Rodde, N., et al., 2023b. The genomics of linkage drag in inbred lines of sunflower. *Proc. Natl. Acad. Sci. U. S. A.* 120.
- Hübner, S., Bercovich, N., Todesco, M., Mandel, J.R., Odenheimer, J., Ziegler, E., Lee, J. S., Baute, G.J., Owens, G.L., Grassa, C.J., et al., 2019. Sunflower pan-genome analysis shows that hybridization altered gene content and disease resistance. *Nat. Plants* 5.
- Huss, C.P., Holmes, K.D., Blubaugh, C.K., 2022. Benefits and Risks of Intercropping for Crop Resilience and Pest Management. *J. Econ. Entomol.* 115.
- Ide, H., Price, W.S., Arata, Y., Ishikawa, M., 1998. Freezing behaviors in leaf buds of cold-hardy conifers visualized by NMR microscopy. *Tree Physiol.* 18.
- Iqbal, Z., Iqbal, M.S., Ahmad, A., Memon, A.G., Ansari, M.I., 2020. New prospects on the horizon: Genome editing to engineer plants for desirable traits. *Curr. Plant Biol.* 24.
- Iqbal, N., Khan, N.A., Ferrante, A., Trivellini, A., Francini, A., Khan, M.I.R., 2017. Ethylene role in plant growth, development and senescence: interaction with other phytohormones. *Front. Plant Sci.* 8.
- Jagadish, S.V.K., 2020. Heat stress during flowering in cereals – effects and adaptation strategies. *New Phytologist* 226.
- Jansen, M.J., Jansma, S.Y., Kuipers, K.M., Vriezen, W.H., Millenaar, F.F., Montoro, T., de Kovel, C.G.F., van Eeuwijk, F.A., Visser, E.J.W., Rieu, I., 2024. Flower bud cooling protects pollen development and improves fertility during heatwaves. *bioRxiv*. 2024.08.18.608441.
- du Jardin, P., 2015. Plant biostimulants: Definition, concept, main categories and regulation. *Scientia Horticulturae* 196.
- Ji, H., Xiao, L., Xia, Y., Song, H., Liu, B., Tang, L., Cao, W., Zhu, Y., Liu, L., 2017. Effects of jointing and booting low temperature stresses on grain yield and yield components in wheat. *Agric. For. Meteorol.* 243.
- Jiang, Y., Lahlali, R., Karunakaran, C., Warkentin, T.D., Davis, A.R., Bueckert, R.A., 2019. Pollen, ovules, and pollination in pea: Success, failure, and resilience in heat. *Plant Cell and Environment* 42.
- Johannes, F., Porcher, E., Teixeira, F.K., Saliba-Colombani, V., Simon, M., Agier, N., Bulski, A., Albuissou, J., Heredia, F., Audigier, P., et al., 2009. Assessing the impact of transgenerational epigenetic variation on complex traits. *PLoS. Genet.* 5.
- Juventia, S.D., Selin Norén, L.L.M., van Apeldoorn, D.F., Ditzler, L., Rossing, W.A.H., 2022. Spatio-temporal design of strip cropping systems. *Agricultural Systems* 201.
- Kamiński, C., Soininen, J.P., Taumberger, M., Dantas, R., Toscano, A., Cinotti, T.S., Maia, R.F., Neto, A.T., 2019. Smart water management platform: IoT-based precision irrigation for agriculture. *Sensors (Switzerland)* 19.
- Katanić, Z., Mlinarić, S., Katanić, N., Čosić, J., Španić, V., 2021. Photosynthetic efficiency in flag leaves and ears of winter wheat during fusarium head blight infection. *Agronomy* 11.
- Kaur, G., Kumar, S., Nayyar, H., Upadhyaya, H.D., 2008. cold stress injury during the pod-filling phase in chickpea (*Cicer arietinum* L.): Effects on quantitative and qualitative components of seeds. *J. Agron. Crop Sci.* 194.
- Kelliher, T., Walbot, V., 2012. Hypoxia triggers meiotic fate acquisition in maize. *Science* (1979) 337.
- Kenen, G., Bekele, E., Imtiaz, M., Dagne, K., 2012. Genetic Vulnerability of Modern Crop Cultivars: Causes, Mechanism and Remedies. *International Journal of Plant Research* 2.
- Khaki, S., Wang, L., 2019. Crop yield prediction using deep neural networks. *Front. Plant Sci.* 10.
- Khan, M.A., Abdullah, Z., 2003. Salinity-sodicity induced changes in reproductive physiology of rice (*Oryza sativa*) under dense soil conditions. *Environ. Exp. Bot.* 49.
- KHATUN, S., TJ, FLOWERS, 1995. Effects of salinity on seed set in rice. *Plant Cell Environ.* 18.
- Kidokoro, S., Shinozaki, K., Yamaguchi-Shinozaki, K., 2022. Transcriptional regulatory network of plant cold-stress responses. *Trends. Plant Sci.* 27.
- Kim, J.M., To, T.K., Matsui, A., Tanoi, K., Kobayashi, N.I., Matsuda, F., Habu, Y., Ogawa, D., Sakamoto, T., Matsunaga, S., et al., 2017. Acetate-mediated novel survival strategy against drought in plants. *Nat. Plants*.
- Klap, C., Yeshayahu, E., Bolger, A.M., Arazi, T., Gupta, S.K., Shabtai, S., Usadel, B., Salts, Y., Barg, R., 2017. Tomato facultative parthenocarp results from SLGAMOUS-LIKE 6 loss of function. *Plant Biotechnol. J.* 15.
- Konecny, T., Nikoghosyan, M., Binder, H., 2023. Machine learning extracts marks of thiamine's role in cold acclimation in the transcriptome of *Vitis vinifera*. *Front. Plant Sci.* 14.
- Korres, N.E., Burgos, N.R., Travlos, I., Vurro, M., Gitsopoulos, T.K., Varanasi, V.K., Duke, S.O., Kudsk, P., Brabham, C., Rouse, C.E., et al., 2019. New directions for integrated weed management: Modern technologies, tools and knowledge discovery. *Advances in Agronomy*.

- Krug, A.S., Drummond, E.B.M., Van Tassel, D.L., Warschefsky, E.J., 2023. The next era of crop domestication starts now. In: *Proceedings of the National Academy of Sciences of the United States of America*, 120.
- Krzyszton, M., Sacharowski, S.P., Manjunath, V.H., Muter, K., Bokota, G., Wang, C., Plewczynski, D., Dobisova, T., Swiezewski, S., 2024. Dormancy heterogeneity among *Arabidopsis thaliana* seeds is linked to individual seed size. *Plant Commun.* 5.
- Kumar, S.P.J., Susmita, C., Sripathy, K.V., Agarwal, D.K., Pal, G., Singh, A.N., Kumar, S., Rai, A.K., Simal-Gandara, J., 2022. Molecular characterization and genetic diversity studies of Indian soybean (*Glycine max* (L.) Merr.) cultivars using SSR markers. *Mol. Biol. Rep.* 49.
- Kurek, I., Chang, T.K., Bertain, S.M., Madrigal, A., Liu, L., Lassner, M.W., Zhu, G., 2007. Enhanced Thermostability of *Arabidopsis* Rubisco Activase Improves Photosynthesis and Growth Rates under Moderate Heat Stress. *Plant Cell* 19, 3230–3241.
- Langan, P., Bernád, V., Walsh, J., Henchy, J., Khodaeiaminjan, M., Mangina, E., Negrão, S., 2022. Phenotyping for waterlogging tolerance in crops: current trends and future prospects. *J. Exp. Bot.* 73.
- Lange, T., Krämer, C., Lange, M.J.P., 2020. The Class III Gibberellin 2-Oxidases AtGA2ox9 and AtGA2ox10 Contribute to Cold Stress Tolerance and Fertility1 [OPEN]. *Plant Physiol.* 184.
- Lazaridi, E., Kapazoglou, A., Gerakari, M., Klefogianni, K., Passa, K., Sarri, E., Papasotiropoulos, V., Tani, E., Bebeli, P.J., 2024. Crop Landraces and Indigenous Varieties: A Valuable Source of Genes for Plant Breeding. *Plants* 13.
- Lee, H., 2018. Stem Cell Maintenance and Abiotic Stress Response in Shoot Apical Meristem for Developmental Plasticity. *Journal of Plant Biology* 61.
- Lesjak, J., Calderini, D.F., 2017. Increased night temperature negatively affects grain yield, biomass and grain number in Chilean quinoa. *Front. Plant Sci.* 8.
- Li, X.M., Chao, D.Y., Wu, Y., Huang, X., Chen, K., Cui, L.G., Su, L., Ye, W.W., Chen, H., Chen, H.C., et al., 2015. Natural alleles of a proteasome $\alpha 2$ subunit gene contribute to thermotolerance and adaptation of African rice. *Nat. Genet.* 47, 827–833.
- Li, H., Hou, X., Du, T., 2023a. Responses of tomato fruit water balance and xylem hydraulic property of pedicel and calyx to water deficit and salinity stress. *Environ. Exp. Bot.* 206.
- Li, P., Jiang, J., Zhang, G., Miao, S., Lu, J., Qian, Y., Zhao, X., Wang, W., Qiu, X., Zhang, F., et al., 2023b. Integrating GWAS and transcriptomics to identify candidate genes conferring heat tolerance in rice. *Front. Plant Sci.* 13.
- Li, Z., Zhou, X., Cheng, Q., Zhai, W., Mao, B., Li, Y., Chen, Z., 2023c. An integrated feature selection approach to high water stress yield prediction. *Front. Plant Sci.* 14.
- Lim, E.C., Kim, J., Park, J., Kim, E.J., Kim, J., Park, Y.M., Cho, H.S., Byun, D., Henderson, I.R., Copenhaver, G.P., et al., 2020. DeepTetrad: high-throughput image analysis of meiotic tetrads by deep learning in *Arabidopsis thaliana*. *Plant Journal* 101.
- Lindsay, P., Swentowsky, K.W., Jackson, D., 2024. Cultivating potential: Harnessing plant stem cells for agricultural crop improvement. *Mol. Plant* 17.
- Liu, D., He, Y., Wang, Y., Chen, W., Yang, J., Zhang, Y., Feng, Y., Zhao, Y., Lin, S., Huang, L., 2024. Tetrad stage transient cold stress skews auxin-mediated energy metabolism balance in Chinese cabbage pollen. *Plant Physiol.* 195.
- Liu, F., Jensen, C.R., Andersen, M.N., 2005. A Review of Drought Adaptation in Crop plants: Changes in Vegetative and Reproductive Physiology Induced By ABA-based Chemical Signals. *Australian Journal of Agricultural Research*.
- Liu, W., Li, R.J., Han, T.T., Cai, W., Fu, Z.W., Lu, Y.T., 2015. Salt stress reduces root meristem size by nitric oxidemediated modulation of auxin accumulation and signaling in *Arabidopsis*. *Plant Physiol.* 168.
- Liu, F., Li, B.H., Lian, S., Dong, X.L., Wang, C.X., Zhang, Z.F., Liang, W.X., 2018. Effects of temperature and moisture on the infection and development of apple fruit rot caused by *Phytophthora cactorum*. *Plant Dis.* 102.
- Liu, L., Zhang, J., Xu, J., Li, Y., Guo, L., Wang, Z., Zhang, X., Zhao, B., Guo, Y.D., Zhang, N., 2020. CRISPR/Cas9 targeted mutagenesis of SLBD40, a lateral organ boundaries domain transcription factor, enhances drought tolerance in tomato. *Plant Science* 301.
- Lohani, N., Singh, M.B., Bhalla, P.L., 2020. High temperature susceptibility of sexual reproduction in crop plants. *J. Exp. Bot.* 71.
- Lukasewski, A.J., Alberti, A., Sharpe, A., Kilian, A., Stanca, A.M., Keller, B., Clavijo, B.J., Friebe, B., Gill, B., Wulff, B., et al., 2014. A chromosome-based draft sequence of the hexaploid bread wheat (*Triticum aestivum*) genome. *Science* (1979) 345.
- Lv, X., Yao, Q., Mao, F., Liu, M., Wang, Y., Wang, X., Gao, Y., Wang, Y., Liao, S., Wang, P., et al., 2024. Heat stress and sexual reproduction in maize: unveiling the most pivotal factors and the greatest opportunities. *J. Exp. Bot.* 75.
- Ma, Y., Min, L., Wang, J., Li, Y., Wu, Y., Hu, Q., Ding, Y., Wang, M., Liang, Y., Gong, Z., et al., 2021. A combination of genome-wide and transcriptome-wide association studies reveals genetic elements leading to male sterility during high temperature stress in cotton. *New Phytologist* 231.
- Máková, K., Prabhullachandran, U., Štefková, M., Spyroglou, I., Pěnčík, A., Endlová, L., Novák, O., Robert, H.S., 2022. Long-Term High-Temperature Stress Impacts on Embryo and Seed Development in *Brassica napus*. *Front. Plant Sci.* 13.
- Manik, S.M.N., Quamruzzaman, M., Livermore, M., Zhao, C., Johnson, P., Hunt, I., Shabala, S., Zhou, M., 2022. Impacts of barley root cortical aerenchyma on growth, physiology, yield components, and grain quality under field waterlogging conditions. *Field. Crops. Res.* 279.
- Maple, R., Zhu, P., Hepworth, J., Wang, J.W., Dean, C., 2024. Flowering time: From physiology, through genetics to mechanism. *Plant Physiol.* 195.
- Martínez-Lorente, S.E., Martí-Guillén, J.M., Pedreño, M.Á., Almagro, L., Sabater-Jara, A. B., 2024. Higher Plant-Derived Biostimulants: Mechanisms of Action and Their Role in Mitigating Plant Abiotic Stress. *Antioxidants* 13.
- Mascher, M., Schreiber, M., Scholz, U., Graner, A., Reif, J.C., Stein, N., 2019. Genebank genomics bridges the gap between the conservation of crop diversity and plant breeding. *Nat. Genet.* 51.
- Mattioli, R., Palombi, N., Funck, D., Trovato, M., 2020. Proline Accumulation in Pollen Grains as Potential Target for Improved Yield Stability Under Salt Stress. *Front. Plant Sci.* 11.
- Mayer, K.F.X., Waugh, R., Langridge, P., Close, T.J., Wise, R.P., Graner, A., Matsumoto, T., Sato, K., Schulman, A., Ariyadasa, R., et al., 2012. A physical, genetic and functional sequence assembly of the barley genome. *Nature* 491.
- Mbow, C., Van Noordwijk, M., Luedeling, E., Neufeldt, H., Minang, P.A., Kowero, G., 2014. Agroforestry solutions to address food security and climate change challenges in Africa. *Curr. Opin. Environ. Sustain.* 6.
- McNellie, J.P., Chen, J., Li, X., Yu, J., 2018. Genetic mapping of foliar and tassel heat stress tolerance in maize. *Crop. Sci.* 58.
- Mendes, M.A., Guerra, R.F., Castelnuovo, B., Silva-Velazquez, Y., Morandini, P., Manrique, S., Baumann, N., Groß-Hardt, R., Dickinson, H., Colombo, L., 2016. Live and let die: A REM complex promotes fertilization through synergic cell death in *Arabidopsis*. *Development (Cambridge)* 143.
- Miller, G., Beery, A., Singh, P.K., Wang, F., Zelingher, R., Motenko, E., Lieberman-Lazarovich, M., 2021. Contrasting processing tomato cultivars unlink yield and pollen viability under heat stress. *AoB Plants.* 13.
- Mishra, B., Kumar, N., Liu, J., Pajeroska-Mukhtar, K.M., 2021. Dynamic Regulatory Event Mining by iDREM in Large-Scale Multi-omics Datasets During Biotic and Abiotic Stress in Plants. *Methods in Molecular Biology*.
- Miyazaki, M., Araki, M., Okamura, K., Ishibashi, Y., Yuasa, T., Iwaya-Inoue, M., 2013. Assimilate translocation and expression of sucrose transporter, OsSUT1, contribute to high-performance ripening under heat stress in the heat-tolerant rice cultivar Genkikusushi. *J. Plant Physiol.* 170.
- Mohapatra, U., Singh, A., Ravikumar, R.L., 2020. Correction to: Effect of gamete selection in improving of heat tolerance as demonstrated by shift in allele frequency in maize (*Zea mays* L.) (*Euphytica*, (2020), 216, 5, (76), 10.1007/s10681-020-02603-z). *Euphytica* 216.
- Monsele, S.P., 2018. *Handbook of Fruit Set and Development*.
- Monteiro, A., Santos, S., 2022. Sustainable Approach to Weed Management: The Role of Precision Weed Management. *Agronomy* 12.
- OA, Montesinos-López, Herr, A.W., Crossa, J., Carter, A.H., 2023. Genomics combined with UAS data enhances prediction of grain yield in winter wheat. *Front. Genet.* 14.
- Moslemi, A., Ades, P.K., Groom, T., Nicolas, M.E., Taylor, P.W.J., 2018. Influence of waterlogging on growth of pyrethrum plants infected by the crown and root rot pathogens, *Fusarium oxysporum*, *Fusarium avenaceum* and *Paraphoma vineae*. *Australasian Plant Pathology* 47.
- Mousavi, S.R., Jahandideh Mahjenabadi, V.A., Khoshru, B., Rezaei, M., 2023. Spatial prediction of winter wheat yield gap: agro-climatic model and machine learning approaches. *Front. Plant Sci.* 14.
- Munns, R., Day, D.A., Fricke, W., Watt, M., Arsova, B., Barkla, B.J., Bose, J., Byrt, C.S., Chen, Z.H., Foster, K.J., et al., 2020. Energy costs of salt tolerance in crop plants. *New Phytologist* 225.
- Munns, R., Tester, M., 2008. Mechanisms of salinity tolerance. *Annu. Rev. Plant Biol.* 59.
- Murmu, S., Sinha, D., Chaurasia, H., Sharma, S., Das, R., Jha, G.K., Archak, S., 2024. A review of artificial intelligence-assisted omics techniques in plant defense: current trends and future directions. *Front. Plant Sci.* 15.
- N Govachelvan, K., Pathmanathan, D., Zainal-Abidin, R.A., Abu, A., 2024. Machine learning for major food crops breeding: Applications, challenges, and ways forward. *Agron.* J. 116.
- Nakano, H., Iwasawa, N., Takai, T., Arai-Sanoh, Y., Kondo, M., 2021. Grain weight and the concentrations of phenylpropanoid glycosides and γ -oryzanol in response to heat stress during ripening in rice. *Cereal. Chem.* 98.
- Nayyar, H., Kaur, G., Kumar, S., Upadhyaya, H.D., 2007. Low temperature effects during seed filling on chickpea genotypes (*Cicer arietinum* L.): Probing mechanisms affecting seed reserves and yield. *J. Agron. Crop. Sci.* 193.
- Neumann Andersen, M., Asch, F., Wu, Y., Richardt Jensen, C., Næsted, H., Overgaard Mogensen, V., Elaine Koch, K., 2002. Soluble invertase expression is an early target of drought stress during the critical, abortion-sensitive phase of young ovary development in maize. *Plant Physiol.* 130.
- Niedbala, G., Piekutowska, M., Wojciechowski, T., Niazian, M., 2024. Predictions and Estimations in Agricultural Production under a Changing Climate. *Agronomy* 14.
- Niinemets, Ü., 2020. Leaf trait plasticity and evolution in different plant functional types. *Annual Plant Reviews Online* 3.
- Ocvirk, D., Špoljarević, M., Kristić, M., Hancock, J.T., Teklić, T., Lisjak, M., 2021. The effects of seed priming with sodium hydrosulphide on drought tolerance of sunflower (*Helianthus annuus* L.) in germination and early growth. *Annals of Applied Biology* 178.
- Oliver, S.N., Dennis, E.S., Dolferus, R., 2007. ABA regulates apoplastic sugar transport and is a potential signal for cold-induced pollen sterility in rice. *Plant and Cell Physiology* 48.
- Oliver, S.N., Van Dongen, J.T., Alfred, S.C., Mamun, E.A., Zhao, X., Saini, H.S., Fernandes, S.F., Blanchard, C.L., Sutton, B.G., Geigenberger, P., et al., 2005. Cold-induced repression of the rice anther-specific cell wall invertase gene OSINV4 is correlated with sucrose accumulation and pollen sterility. *Plant, Cell and Environment* 28.
- Pan, J., Sharif, R., Xu, X., Chen, X., 2021. Mechanisms of Waterlogging Tolerance in Plants: Research Progress and Prospects. *Front. Plant Sci.* 11.
- Pan, F., Wu, M., Hu, W., Liu, R., Yan, H., Xiang, Y., 2019a. Genome-wide identification and expression analyses of the bZIP transcription factor genes in moso bamboo (*Phyllostachys edulis*). *Int. J. Mol. Sci.* 20, 1–21.
- Pan, C., Yang, D., Zhao, X., Jiao, C., Yan, Y., Lamin-Samu, A.T., Wang, Q., Xu, X., Fei, Z., Lu, G., 2019b. Tomato stigma exertion induced by high temperature is associated with the jasmonate signalling pathway. *Plant Cell and Environment* 42.

- Panagos, P., Matthews, F., Patault, E., De Michele, C., Quaranta, E., Bezak, N., Kaffas, K., Patro, E.R., Auel, C., Schleiss, A.J., et al., 2024. Understanding the cost of soil erosion: An assessment of the sediment removal costs from the reservoirs of the European Union. *J. Clean. Prod.* 434.
- Panahirad, S., Dadpour, M., Gohari, G., Akbari, A., Mahdavinia, G., Jafari, H., Kulak, M., Alcázar, R., Fotopoulos, V., 2023. Putrescine-functionalized carbon quantum dot (put-CQD) nanoparticle: A promising stress-protecting agent against cadmium stress in grapevine (*Vitis vinifera* cv. Sultan). *Plant Physiology and Biochemistry* 197.
- Pandey, S., Singh, A., Parida, S.K., Prasad, M., 2022. Combining speed breeding with traditional and genomics-assisted breeding for crop improvement. *Plant Breeding* 141.
- Pantera, Mosquera-Losada MR, Herzog, F., den Herder, M., 2021. Agroforestry and the environment. *Agrofor. Syst.* 95.
- Parrotta, L., Faleri, C., Guerriero, G., Cai, G., 2019. Cold stress affects cell wall deposition and growth pattern in tobacco pollen tubes. *Plant Science* 283.
- Parry, M., Rosenzweig, C., Livermore, M., 2005. Climate change, global food supply and risk of hunger. *Philosophical Transactions of the Royal Society B: Biological Sciences*.
- Pat-Cetina, J.E., Orozco-del-Castillo, M.G., López-Puerto, K.A., Bermejo-Sabbagh, C., Cuevas-Cuevas, N.L., 2023. Recognition of Pollen-Carrying Bees Using Convolutional Neural Networks and Digital Image Processing Techniques. *Communications in Computer and Information Science*.
- Pazhamala, L.T., Kudapa, H., Weckwerth, W., Millar, A.H., Varshney, R.K., 2021. Systems biology for crop improvement. *Plant Genome* 14.
- Penfield, S., MacGregor, D.R., 2017. Effects of environmental variation during seed production on seed dormancy and germination. *J. Exp. Bot.* 68.
- Penfield, S., Warner, S., Wilkinson, L., 2021. Molecular responses to chilling in a warming climate and their impacts on plant reproductive development and yield. *J. Exp. Bot.* 72.
- Perotti, M.F., Posé, D., Martín-Pizarro, C., 2023. Non-climacteric fruit development and ripening regulation: 'the phytohormones show'. *J. Exp. Bot.* 74.
- Pessoa, H.P., Dariva, F.D., Copati, M.G.F., de Paula, R.G., de Oliveira Dias, F., Gomes, C. N., 2023. Uncovering tomato candidate genes associated with drought tolerance using *Solanum pennellii* introgression lines. *PLoS. One* 18.
- Petersen, K.K., Willumsen, J., Kaack, K., 1998. Composition and taste of tomatoes as affected by increased salinity and different salinity sources. *Journal of Horticultural Science and Biotechnology* 73.
- Picard, M., Scott-Boyer, M.P., Bodein, A., Péron, O., Droit, A., 2021. Integration strategies of multi-omics data for machine learning analysis. *Comput. Struct. Biotechnol. J.* 19.
- Pixley, K.V., Cairns, J.E., Lopez-Ridaura, S., Ojiewo, C.O., Dawud, M.A., Drabo, I., Mindaye, T., Nebie, B., Asea, G., Das, B., et al., 2023. Redesigning crop varieties to win the race between climate change and food security. *Mol. Plant* 16.
- Pizarro, L., Leibman-Markus, M., Gupta, R., Kovetz, N., Shtein, I., Bar, E., Davidovich-Rikanati, R., Zarivach, R., Lewinsohn, E., Avni, A., et al., 2020. A gain of function mutation in *SINRC4a* enhances basal immunity resulting in broad-spectrum disease resistance. *Commun. Biol.* 3.
- Pressman, E., Peet, M.M., Pharr, D.M., 2002. The effect of heat stress on tomato pollen characteristics is associated with changes in carbohydrate concentration in the developing anthers. *Ann. Bot.* 90, 631–636.
- Priya, M., Sharma, L., Kaur, R., Bindumadhava, H., Nair, R.M., Siddique, K.H.M., Nayyar, H., 2019a. GABA (γ -aminobutyric acid), as a thermo-protectant, to improve the reproductive function of heat-stressed mungbean plants. *Sci. Rep.* 9.
- Priya, M., Sharma, L., Singh, I., Bains, T.S., Siddique, K.H.M., H. B., Nair, R.M., Nayyar, H., 2019b. Securing reproductive function in mungbean grown under high temperature environment with exogenous application of proline. *Plant Physiology and Biochemistry* 140.
- Prohens, J., Gramazio, P., Plazas, M., Dempewolf, H., Kilian, B., Díez, M.J., Fita, A., Herranz, F.J., Rodríguez-Burruezo, A., Soler, S., et al., 2017. Introgressomics: a new approach for using crop wild relatives in breeding for adaptation to climate change. *Euphytica* 213.
- Qi, Z.Y., Wang, K.X., Yan, M.Y., Kanwar, M.K., Li, D.Y., Wijaya, L., Alyemeni, M.N., Ahmad, P., Zhou, J., 2018. Melatonin alleviates high temperature-induced pollen abortion in *solanum lycopersicum*. *Molecules*. 23.
- Qi, B., Wu, C., 2022. Potential roles of stigma exertion on spikelet fertility in rice (*Oryza sativa* L.) under heat stress. *Front. Plant Sci.* 13.
- Qian-Chuan, L., Shi-Wei, X., Jia-Yu, Z., Jia-Jia, L., Yi, Z., Ze-Xi, Z., 2023. Ensemble learning prediction of soybean yields in China based on meteorological data. *Scientia Agricultura Sinica* 22.
- Quandt, A., Neufeldt, H., Gorman, K., 2023. Climate change adaptation through agroforestry: opportunities and gaps. *Curr. Opin. Environ. Sustain.* 60.
- Rai, K.K., 2022. Integrating speed breeding with artificial intelligence for developing climate-smart crops. *Mol. Biol. Rep.* 49.
- Rai, A., Yamazaki, M., Saito, K., 2019. A new era in plant functional genomics. *Curr. Opin. Syst. Biol.* 15.
- Raihan, T., Geneve, R.L., Perry, S.E., Lopez, C.M.R., 2021. The regulation of plant vegetative phase transition and rejuvenation: Mirnas, a key regulator. *Epigenomes*. 5.
- Rao, M.J., Duan, M., Shad, M.A., Aslam, M.Z., Wang, J., Wang, L., 2024. Widely targeted LC-MS/MS approach provides insights into variations in bioactive flavonoid compounds and their antioxidant activities in green, red, and purple sugarcane. *Lwt* 209, 116792.
- Raza, A., Charagh, S., Najafi-Kakavand, S., Abbas, S., Shoaib, Y., Anwar, S., Sharifi, S., Lu, G., Siddique, K.H.M., 2023. Role of phytohormones in regulating cold stress tolerance: Physiological and molecular approaches for developing cold-smart crop plants. *Plant Stress* 8.
- Raza, Q., Riaz, A., Bashir, K., Sabar, M., 2020. Reproductive tissues-specific meta-QTLs and candidate genes for development of heat-tolerant rice cultivars. *Plant Mol. Biol.* 104.
- Reinders, J., Wulff, B.B.H., Mirouze, M., Mari-Ordóñez, A., Dapp, M., Rozhon, W., Bucher, E., Theiler, G., Paszkowski, J., 2009. Compromised stability of DNA methylation and transposon immobilization in mosaic Arabidopsis epigenomes. *Genes and Development* 23.
- Rezaul, I.M., Baohua, F., Tingting, C., Weimeng, F., Caixia, Z., Longxing, T., Guanfu, F., 2019. Absciscic acid prevents pollen abortion under high-temperature stress by mediating sugar metabolism in rice spikelets. *Plant* 165.
- Riaz, A., Periyannan, S., Aitken, E., Hickey, L., 2016. A rapid phenotyping method for adult plant resistance to leaf rust in wheat. *Plant Methods* 12.
- Rieu, I., Twell, D., Firon, N., 2017. Pollen development at high temperature: From acclimation to collapse. *Plant Physiol.* 173, 1967–1976.
- Rivelli, G.M., Calderini, D.F., Abeledo, L.G., Miralles, D.J., Rondonani, D.P., 2024. Yield and quality traits of wheat and rapeseed in response to source-sink ratio and heat stress in post-flowering. *European Journal of Agronomy* 152.
- Rivero, R.M., Mittler, R., Blumwald, E., Zandalinas, S.I., 2022. Developing climate-resilient crops: improving plant tolerance to stress combination. *Plant Journal* 109.
- JF, Roca Paixão, Gillet, F.X., Ribeiro, T.P., Bournaud, C., Lourenço-Tessutti, I.T., Noriega, D.D., de Melo BP, de Almeida-Engler, J., Grossi-de-Sa, M.F., 2019. Improved drought stress tolerance in Arabidopsis by CRISPR/dCas9 fusion with a Histone Acetyltransferase. *Sci. Rep.* 9.
- Rodene, E., Fernando, G.D., Piyush, V., Ge, Y., Schnable, J.C., Ghosh, S., Yang, J., 2024. Image Filtering to Improve Maize Tassel Detection Accuracy Using Machine Learning Algorithms. *Sensors* 24, 1–14.
- Rodrigues, J., Inzé, D., Nelissen, H., Saibo, N.J.M., 2019. Source-Sink Regulation in Crops under Water Deficit. *Trends. Plant Sci.* 24.
- Rohart, F., Eslami, A., Matigian, N., Bougeard, S., Lê Cao, K.A., 2017a. MINT: A multivariate integrative method to identify reproducible molecular signatures across independent experiments and platforms. *BMC. Bioinformatics*. 18.
- Rohart, F., Gautier, B., Singh, A., Lê Cao, K.A., 2017b. mixOmics: An R package for 'omics feature selection and multiple data integration. *PLoS. Comput. Biol.* 13.
- Rouphael, Y., Colla, G., Giordano, M., El-Nakheel, C., Kyriacou, M.C., De Pascale, S., 2017. Foliar applications of a legume-derived protein hydrolysate elicit dose-dependent increases of growth, leaf mineral composition, yield and fruit quality in two greenhouse tomato cultivars. *Scientia Horticulturae* 226.
- Roychowdhury, R., Das, S.P., Gupta, A., Parihar, P., Chandrasekhar, K., Sarker, U., Kumar, A., Ramrao, D.P., Sudhakar, C., 2023. Multi-Omics Pipeline and Omics-Integration Approach to Decipher Plant's Abiotic Stress Tolerance Responses. *Genes. (Basel)* 14.
- Ruan, Y.L., Patrick, J.W., Bouzayen, M., Osorio, S., Fernie, A.R., 2012. Molecular regulation of seed and fruit set. *Trends. Plant Sci.* 17.
- Rymen, B., Fiorani, F., Kartal, F., Vandepoele, K., Inzé, D., Beebe, G.T.S., 2007. Cold nights impair leaf growth and cell cycle progression in maize through transcriptional changes of cell cycle genes. *Plant Physiol.* 143.
- Sanghera, G.H.S., Wani, S., Hussain, W., B Singh, N., 2011. Engineering Cold Stress Tolerance in Crop Plants. *Curr. Genomics*. 12.
- Sadok, W., Lopez, J.R., Smith, K.P., 2021. Transpiration increases under high-temperature stress: Potential mechanisms, trade-offs and prospects for crop resilience in a warming world. *Plant Cell and Environment* 44.
- Sage, T.L., Bagha, S., Lundsgaard-Nielsen, V., Branch, H.A., Sultmanis, S., Sage, R.F., 2015. The effect of high temperature stress on male and female reproduction in plants. *Field. Crops. Res.* 182.
- Salgotra, R.K., Chauhan, B.S., 2023. Genetic Diversity, Conservation, and Utilization of Plant Genetic Resources. *Genes. (Basel)* 14.
- Sallam, A., Alqudah, A.M., Dawood, M.F.A., Baenziger, P.S., Börner, A., 2019. Drought stress tolerance in wheat and barley: Advances in physiology, breeding and genetics research. *Int. J. Mol. Sci.* 20.
- Van de Sande, B., Flerin, C., Davie, K., De Waegeneer, M., Hulselmans, G., Aibar, S., Seurinck, R., Saelens, W., Cannoodt, R., Rouchon, Q., et al., 2020. A scalable SCENIC workflow for single-cell gene regulatory network analysis. *Nat. Protoc.* 15.
- Sandhu, K.S., Lozada, D.N., Zhang, Z., Pumphrey, M.O., Carter, A.H., 2021. Deep Learning for Predicting Complex Traits in Spring Wheat Breeding Program. *Front. Plant Sci.* 11.
- Sati, H., Khandelwal, A., Pareek, S., 2023. Effect of exogenous melatonin in fruit postharvest, crosstalk with hormones, and defense mechanism for oxidative stress management. *Food Front.* 4.
- Saxena, R., Joshi, A., Joshi, S., Borkotoky, S., Singh, K., Rai, P.K., Mueed, Z., Sharma, R., 2022. The role of artificial intelligence strategies to mitigate abiotic stress and climate change in crop production. *Visualization Techniques for Climate Change with Machine Learning and Artificial Intelligence*.
- Scaven, V.L., Rafferty, N.E., 2013. Physiological effects of climate warming on flowering plants and insect pollinators and potential consequences for their interactions. *Curr. Zool.* 59.
- Schmid, M.W., Heichinger, C., Coman Schmid, D., Guthörl, D., Gagliardini, V., Bruggmann, R., Aluri, S., Aquino, C., Schmid, B., Turnbull, L.A., et al., 2018. Contribution of epigenetic variation to adaptation in Arabidopsis. *Nat. Commun.* 9.
- Schmidt, M., Felsche, E., 2024. The effect of climate change on crop yield anomaly in Europe. *Climate Resilience and Sustainability* 3.
- Schnable, P.S., Ware, D., Fulton, R.S., Stein, J.C., Wei, F., Pasternak, S., Liang, C., Zhang, J., Fulton, L., Graves, T.A., et al., 2009. The B73 maize genome: Complexity, diversity, and dynamics. *Science (1979)* 326.
- Seetharam, K., Kuchanur, P.H., Koirala, K.B., Tripathi, M.P., Patil, A., Sudarsanam, V., Das, R.R., Chaurasia, R., Pandey, K., Vemuri, H., et al., 2021. Genomic regions associated with heat stress tolerance in tropical maize (*Zea mays* L.). *Sci. Rep.* 11.

- Sehgal, A., Sita, K., Siddique, K.H.M., Kumar, R., Bhogireddy, S., Varshney, R.K., HanumanthaRao, B., Nair, R.M., Prasad, P.V.V., Nayyar, H., 2018. Drought or/and heat-stress effects on seed filling in food crops: Impacts on functional biochemistry, seed yields, and nutritional quality. *Front. Plant Sci.* 871.
- Shaar-Moshe, L., Hayouka, R., Roessner, U., Peleg, Z., 2019. Phenotypic and metabolic plasticity shapes life-history strategies under combinations of abiotic stresses. *Plant Direct* 3.
- Shaikh, T.A., Mir, W.A., Rasool, T., Sofi, S., 2022. Machine Learning for Smart Agriculture and Precision Farming: Towards Making the Fields Talk. *Archives of Computational Methods in Engineering* 29.
- Shang, L., Song, J., Yu, H., Wang, X., Yu, C., Wang, Y., Li, F., Lu, Y., Wang, T., Ouyang, B., et al., 2021. A mutation in a C2H2-type zinc finger transcription factor contributed to the transition toward self-pollination in cultivated tomato. *Plant Cell* 33, 3293–3308.
- Shao, J., Huang, K., Batool, M., Idrees, F., Afzal, R., Haroon, M., Noushahi, H.A., Wu, W., Hu, Q., Lu, X., et al., 2022. Versatile roles of polyamines in improving abiotic stress tolerance of plants. *Front. Plant Sci.* 13.
- Sharma, A., Hazarika, M., Heisnam, P., Pandey, H., Devadas, V.A.S.N., Kesavan, A.K., Kumar, P., Singh, D., Vashishth, A., Jha, R., et al., 2024. Controlled Environment Ecosystem: A Cutting-Edge Technology in Speed Breeding. *ACS. Omega* 9, 29114–29138.
- Sharma, M., Negi, S., Kumar, P., Srivastava, D.K., Choudhary, M.K., Irfan, M., 2023a. Fruit ripening under heat stress: The intriguing role of ethylene-mediated signaling. *Plant Science* 335.
- Sharma, A., Sharma, A., Tselykh, A., Bozhenyuk, A., Choudhury, T., Alomar, M.A., Sánchez-Chero, M., 2023b. Artificial intelligence and internet of things oriented sustainable precision farming: Towards modern agriculture. *Open. Life Sci.* 18.
- Sheikh, M., Iqra, F., Ambreen, H., Pravin, K.A., Ikra, M., Chung, Y.S., 2024. Integrating artificial intelligence and high-throughput phenotyping for crop improvement. *Journal of Integrative Agriculture* 23.
- Sheng, X., Ai, Z., Tan, Y., Hu, Y., Guo, X., Liu, X., Sun, Z., Yu, D., Chen, J., Tang, N., et al., 2023. Novel Salinity-Tolerant Third-Generation Hybrid Rice Developed via CRISPR/Cas9-Mediated Gene Editing. *Int. J. Mol. Sci.* 24.
- Sheoran, S., 2022. Recent Advances for Drought Stress Tolerance in Maize (*Zea mays* L.): Present Status and Future Prospects. *Front. Plant Sci.* 13.
- Shi, J., Gao, H., Wang, H., Lafitte, H.R., Archibald, R.L., Yang, M., Hakimi, S.M., Mo, H., Habben, J.E., 2017. ARGOS8 variants generated by CRISPR-Cas9 improve maize grain yield under field drought stress conditions. *Plant Biotechnol. J.* 15.
- Shiade, S.R.G., Zand-Silakhoor, A., Fathi, A., Rahimi, R., Minkina, T., Rajput, V.D., Zulfikar, U., Chaudhary, T., 2024. Plant metabolites and signaling pathways in response to biotic and abiotic stresses: Exploring bio stimulant applications. *Plant Stress* 12, 100454.
- Simontacchi, M., Galatrol, A., Ramos-Arturo, F., GE, Santa-María, 2015. Plant survival in a changing environment: The role of nitric oxide in plant responses to abiotic stress. *Front. Plant Sci.* 6.
- Singh, A., 2022. Soil salinity: A global threat to sustainable development. *Soil. Use Manage* 38.
- Singh, A., Antre, S.H., Ravikumar, R.L., Kuchanur, P.H., Lohithaswa, H.C., 2020. Genetic evidence of pollen selection mediated phenotypic changes in maize conferring transgenerational heat-stress tolerance. *Crop. Sci.* 60.
- Singh, B.K., Delgado-Baquerizo, M., Egidi, E., Guirado, E., Leach, J.E., Liu, H., Trivedi, P., 2023. Climate change impacts on plant pathogens, food security and paths forward. *Nature Reviews Microbiology* 21.
- Singh, A., Jones, S., Ganapathysubramanian, B., Sarkar, S., Mueller, D., Sandhu, K., Nagasubramanian, K., 2021a. Challenges and Opportunities in Machine-Augmented Plant Stress Phenotyping. *Trends. Plant Sci.* 26.
- Singh, P.K., Miller, G., Faigenboim, A., Lieberman-Lazarovich, M., 2021. The tomato ddm1b mutant shows decreased sensitivity to heat stress accompanied by transcriptional alterations. *Genes* 12 (9), 1337.
- Singh, S., Singh, A.V., Singh, A., Singh, S.P., 2021c. Speed Breeding: An Innovative Method for Crop Improvement. *EC Agriculture*.
- Skobalski, J., Sagan, V., Alifu, H., Al Akkad, O., Lopes, F.A., Grignola, F., 2024. Bridging the gap between crop breeding and GeoAI: Soybean yield prediction from multispectral UAV images with transfer learning. *ISPRS Journal of Photogrammetry and Remote Sensing* 210, 260–281.
- Snider, J.L., Oosterhuis, D.M., Skulman, B.W., Kawakami, E.M., 2009. Heat stress-induced limitations to reproductive success in *Gossypium hirsutum*. *Physiol. Plant* 137.
- Sosnowski, J., Truba, M., Vasileva, V., 2023. The Impact of Auxin and Cytokinin on the Growth and Development of Selected Crops. *Agriculture (Switzerland)* 13.
- Soualio, S., Duan, F., Li, X., Zhou, W., 2022. CROP PRODUCTION UNDER COLD STRESS: An understanding of plant responses, acclimation processes, and management strategies. *Plant Physiology and Biochemistry* 190.
- Spanic, V., Korzun, V., Ebmeyer, E., 2016. Assessing genetic diversity of wheat genotypes from different origins by SNP markers. *Cereal. Res. Commun.* 44.
- Španić, V., Šunić, K., Duvnjak, J., Hu, Y.G., Katanić, Z., 2023. Chlorophyll a fluorescence during flag leaf senescence of field-grown winter wheat plants under drought conditions. *Annals of Applied Biology* 183.
- Srinivasan, A., Saxena, N.P., Johansen, C., 1999. Cold tolerance during early reproductive growth of chickpea (*Cicer arietinum* L.): Genetic variation in gamete development and function. *Field. Crops. Res.* 60.
- The State of Food Security and Nutrition in the World 2022. 2022.**
- Stefanski, P., Ullah, S., Matysik, P., Rybka, K., 2024. Triticale field phenotyping using RGB camera for ear counting and yield estimation. *J. Appl. Genet.* 65.
- Stenberg, J.A., 2017. A Conceptual Framework for Integrated Pest Management. *Trends. Plant Sci.* 22.
- Su, Z., Ma, X., Guo, H., Sukiran, N.L., Guo, B., Assmann, S.M., Ma, H., 2013. Flower development under drought stress: Morphological and transcriptomic analyses reveal acute responses and long-term acclimation in *Arabidopsis*. *Plant Cell* 25.
- Suwa, R., Hakata, H., Hara, H., El-Shemy, H.A., Adu-Gyamfi, J.J., Nguyen, N.T., Kanai, S., Lightfoot, D.A., Mohapatra, P.K., Fujita, K., 2010. High temperature effects on photosynthate partitioning and sugar metabolism during ear expansion in maize (*Zea mays* L.) genotypes. *Plant Physiology and Biochemistry* 48.
- Sze, H., Klodová, B., Ward, J.M., Harper, J.F., Palanivelu, R., Johnson, M.A., Honys, D., 2024. A wave of specific transcript and protein accumulation accompanies pollen dehydration. *Plant Physiol.* 195, 1775–1795.
- Taghavi Namin, S., Esmaeilzadeh, M., Najafi, M., Brown, T.B., Borevitz, J.O., 2018. Deep phenotyping: Deep learning for temporal phenotype/genotype classification. *Plant Methods* 14.
- Tan, Z., Yang, J., Li, Q., Su, F., Yang, T., Wang, W., Aierxi, A., Zhang, X., Yang, W., Kong, J., et al., 2022. PollenDetect: An Open-Source Pollen Viability Status Recognition System Based on Deep Learning Neural Networks. *Int. J. Mol. Sci.* 23.
- Tang, J., Tian, X., Mei, E., He, M., Gao, J., Yu, J., Xu, M., Liu, J., Song, L., Li, X., et al., 2022. WRKY53 negatively regulates rice cold tolerance at the booting stage by fine-tuning another gibberellin levels. *Plant Cell* 34.
- Thakur, P., Kumar, S., Malik, J.A., Berger, J.D., Nayyar, H., 2010. Cold stress effects on reproductive development in grain crops: An overview. *Environ. Exp. Bot.* 67.
- Thakur, P., Kumari, N., Kumar, A., Sharma, P., Chadha, S., 2024. Recent advances in development and utilization of double haploids (DHs) in economically important vegetable crops. *Plant Cell Tissue Organ. Cult.* 156.
- Thomashow, M.F., 1999. Plant cold acclimation: Freezing tolerance genes and regulatory mechanisms. *Annu. Rev. Plant Biol.* 50.
- Tong, C., Hill, C.B., Zhou, G., Zhang, X.Q., Jia, Y., Li, C., 2021. Opportunities for improving waterlogging tolerance in cereal crops—Physiological traits and genetic mechanisms. *Plants* 10.
- Tong, H., Nikoloski, Z., 2021. Machine learning approaches for crop improvement: Leveraging phenotypic and genotypic big data. *J. Plant Physiol.* 257.
- Totsky, I.V., Lyakh, V.A., 2015. Pollen selection for drought tolerance in sunflower. *Helia* 38.
- Toulote, J.M., Pantazopoulou, C.K., Sanclemente, M.A., Voesenek, L.A.C.J., Sasiidharan, R., 2022. Water stress resilient cereal crops: Lessons from wild relatives. *J. Integr. Plant Biol.* 64.
- Tsuji, H., Saika, H., Tsutsumi, N., Hirai, A., Nakazono, M., 2006. Dynamic and reversible changes in histone H3-Lys4 methylation and H3 acetylation occurring at submergence-inducible genes in rice. *Plant and Cell Physiology* 47.
- Turc, O., Tardieu, F., 2018. Drought affects abortion of reproductive organs by exacerbating developmentally driven processes via expansive growth and hydraulics. *J. Exp. Bot.* 69.
- Tyagi, A., Ali, S., Park, S., Bae, H., 2023. Exploring the Potential of Multiomics and Other Integrative Approaches for Improving Waterlogging Tolerance in Plants. *Plants* 12.
- Ullah, F., Saqib, S., Khan, W., Zhao, L., Khan, W., Li, M.Y., Xiong, Y.C., 2024. Sodium Nitroprusside and Melatonin Improve Physiological Vitality and Drought Acclimation via Synergistically Enhancing Antioxidant Response in Dryland Maize. *J. Plant Growth Regul.*
- Umashaanker, M., Solomon, D., Rao, P., Singh, S., Ishtiaque, A., Balwinder-Singh, Poonia S., Agrawal, A., Jain, M., 2024. Groundwater irrigation is critical for adapting wheat systems to warming temperatures in the Eastern Indo-Gangetic Plains in India. *Environmental Research: Food Systems* 1, 021002.
- Underwood, C.J., Mercier, R., 2022. Engineering Apomixis: Clonal Seeds Approaching the Fields. *Annu. Rev. Plant Biol.* 73.
- Uri, N., 2018. Cropland soil salinization and associated hydrology: Trends, processes and examples. *Water (Switzerland)* 10.
- Usman, B., Nawaz, G., Zhao, N., Liao, S., Liu, Y., Li, R., 2020. Precise editing of the *ospyl9* gene by rna-guided cas9 nuclease confers enhanced drought tolerance and grain yield in rice (*Oryza sativa* L.) by regulating circadian rhythm and abiotic stress responsive proteins. *Int. J. Mol. Sci.* 21.
- Uygun, S., Azodi, C.B., Shiu, S.H., 2019. Cis-regulatory code for predicting plant cell-type transcriptional response to high salinity. *Plant Physiol.* 181.
- Varala, K., Marshall-Colón, A., Cirrone, J., Brooks, M.D., Pasquino, A.V., Lérán, S., Mittal, S., Rock, T.M., Edwards, M.B., Kim, G.J., et al., 2018. Temporal transcriptional logic of dynamic regulatory networks underlying nitrogen signaling and use in plants. In: *Proceedings of the National Academy of Sciences of the United States of America*, 115.
- Varotto, S., Tani, E., Abraham, E., Krugman, T., Kapazoglou, A., Melzer, R., Radanović, A., Miladinović, D., 2020. Epigenetics: Possible applications in climate-smart crop breeding. *J. Exp. Bot.* 71.
- Verma, S., Attuluri, V.P.S., Robert, H.S., 2022. Transcriptional control of *Arabidopsis* seed development. *Planta* 255.
- De Vries, F.T., Griffiths, R.I., Knight, C.G., Nicolitch, O., Williams, A., 2020. Harnessing rhizosphere microbiomes for drought-resilient crop production. *Science (1979)* 368.
- Walck, J.L., Hidayati, S.N., Dixon, K.W., Thompson, K., Poschold, P., 2011. Climate change and plant regeneration from seed. *Glob. Chang. Biol.* 17.
- Walker, C.H., Bennett, T., 2024. Cytokinin and reproductive shoot architecture: bigger and better? *Biochem. Soc. Trans.* 52, 1885–1893.
- Walsh, J.J., Mangina, E., Negrão, S., 2024. Advancements in Imaging Sensors and AI for Plant Stress Detection: A Systematic Literature Review. *Plant Phenomics*. 6.
- Waltz, E., 2017. Digital farming attracts cash to agtech startups. *Nat. Biotechnol.* 35.
- Wang, Y., Impa, S.M., Sunkar, R., Jagadish, S.V.K., 2021a. The neglected other half - role of the pistil in plant heat stress responses. *Plant Cell and Environment* 44.
- Wang, C.J.R., Nan, G.L., Kelliher, T., Timofejeva, L., Vernoud, V., Golubovskaya, I.N., Harper, L., Egger, R., Walbot, V., Zacheus Cande, W., 2012. Maize multiple

- archesporial cells 1 (mac1), an ortholog of rice TDL1A, modulates cell proliferation and identity in early anther development. *Development* (Cambridge) 139.
- Wang, W., Wang, X., Zhang, J., Huang, M., Cai, J., Zhou, Q., Dai, T., Jiang, D., 2020. Salicylic acid and cold priming induce late-spring freezing tolerance by maintaining cellular redox homeostasis and protecting photosynthetic apparatus in wheat. *Plant Growth Regul.* 90.
- Wang, W., Wang, X., Zhang, X., Wang, Y., Huo, Z., Huang, M., Cai, J., Zhou, Q., Jiang, D., 2021b. Involvement of salicylic acid in cold priming-induced freezing tolerance in wheat plants. *Plant Growth Regul.* 93.
- Webber, H., Ewert, F., Olesen, J.E., Müller, C., Fronzek, S., Ruane, A.C., Bourgault, M., Martre, P., Ababaei, B., Bindi, M., et al., 2018. Diverging importance of drought stress for maize and winter wheat in Europe. *Nat. Commun.* 9.
- Weckwerth, W., Ghatak, A., Bellaire, A., Chaturvedi, P., Varshney, R.K., 2020. . PANOMICS meets germplasm. *Plant Biotechnol. J.* 18.
- Wen, J., Jiang, F., Weng, Y., Sun, M., Shi, X., Zhou, Y., Yu, L., Wu, Z., 2019. Identification of heat-tolerance QTLs and high-temperature stress-responsive genes through conventional QTL mapping, QTL-seq and RNA-seq in tomato. *BMC. Plant Biol.* 19.
- Wilcock, C., Neiland, R., 2002. Pollination failure in plants: Why it happens and when it matters. *Trends. Plant Sci.* 7.
- Wu, M., Bian, X., Huang, B., Du, Y., Hu, S., Wang, Y., Shen, J., Wu, S., 2024. HD-Zip proteins modify floral structures for self-pollination in tomato. *Science* (1979) 384, 124–130.
- Wu, C., Cui, K., Hu, Q., Wang, W., Nie, L., Huang, J., Peng, S., 2019. Enclosed stigma contributes to higher spikelet fertility for rice (*Oryza sativa* L.) subjected to heat stress. *Crop Journal* 7.
- Wu, C., Cui, K., Wang, W., Li, Q., Fahad, S., Hu, Q., Huang, J., Nie, L., Peng, S., 2016. Heat-induced phytohormone changes are associated with disrupted early reproductive development and reduced yield in rice. *Sci. Rep.* 6.
- Wu, J., Sun, W., Sun, C., Xu, C., Li, S., Li, P., Xu, H., Zhu, D., Li, M., Yang, L., et al., 2023. Cold stress induces malformed tomato fruits by breaking the feedback loops of stem cell regulation in floral meristem. *New Phytologist* 237.
- Xin, X., Li, P., Zhao, X., Yu, Y., Wang, W., Jin, G., Wang, J., Sun, L., Zhang, D., Zhang, F., et al., 2024. Temperature-dependent jumoni demethylase modulates flowering time by targeting H3K36me2/3 in *Brassica rapa*. *Nat. Commun.* 15.
- Xu, J., Driedonks, N., Rutten, M.J.M., Vriezen, W.H., de Boer, G.J., Rieu, I., 2017. Mapping quantitative trait loci for heat tolerance of reproductive traits in tomato (*Solanum lycopersicum*). *Molecular Breeding* 37, 58.
- Xu, K., Xu, X., Fukao, T., Canlas, P., Maghirang-Rodriguez, R., Heuer, S., Ismail, A.M., Bailey-Serres, J., Ronald, P.C., Mackill, D.J., 2006. Sub1A is an ethylene-response-factor-like gene that confers submergence tolerance to rice. *Nature* 442.
- Yakhin, O.I., Lubyantsev, A.A., Yakhin, I.A., Brown, P.H., 2017. Biostimulants in plant science: A global perspective. *Front. Plant Sci.* 7.
- Yamazaki, A., Takezawa, A., Nagasaka, K., Motoki, K., Nishimura, K., Nakano, R., Nakazaki, T., 2023. A simple method for measuring pollen germination rate using machine learning. *Plant Reprod.* 36.
- Yan, J., Wang, X., 2023. Machine learning bridges omics sciences and plant breeding. *Trends. Plant Sci.* 28.
- Yang, X., Kundariya, H., Xu, Y.Z., Sandhu, A., Yu, J., Hutton, S.F., Zhang, M., Mackenzie, S.A., 2015. Muts HOMOLOG1-derived epigenetic breeding potential in tomato. *Plant Physiol.* 168.
- Yocca, A.E., Edger, P.P., 2022. Machine learning approaches to identify core and dispensable genes in pangenomes. *Plant Genome* 15.
- Yoong, F.Y., O'Brien, L.K., Truco, M.J., Huo, H., Sideman, R., Hayes, R., Michellmore, R. W., Bradford, K.J., 2016. Genetic variation for thermotolerance in lettuce seed germination is associated with temperature-sensitive regulation of ethylene response factor1 (ERF1)1[OPEN]. *Plant Physiol.* 170.
- Young, L.W., Wilen, R.W., PC, Bonham-Smith, 2004. High temperature stress of *Brassica napus* during flowering reduces micro- and megagametophyte fertility, induces fruit abortion, and disrupts seed production. *J. Exp. Bot.* 55.
- Yu, H., Lin, T., Meng, X., Du, H., Zhang, J., Liu, G., Chen, M., Jing, Y., Kou, L., Li, X., et al., 2021. A route to de novo domestication of wild allotetraploid rice. *Cell* 184.
- Yu, H., Xie, W., Wang, J., Xing, Y., Xu, C., Li, X., Xiao, J., Zhang, Q., 2011. Gains in QTL detection using an ultra-high density SNP map based on population sequencing relative to traditional RFLP/SSR markers. *PLoS. One* 6.
- Van Zelm, E., Zhang, Y., Testerink, C., 2020. Salt Tolerance Mechanisms of Plants. *Annu Rev. Plant Biol.* 71.
- Zeng, Y., Wen, J., Zhao, W., Wang, Q., Huang, W., 2020. Rational Improvement of Rice Yield and Cold Tolerance by Editing the Three Genes OsPIN5b, GS3, and OsMYB30 With the CRISPR–Cas9 System. *Front. Plant Sci.* 10.
- Zhang, Y.Y., Fischer, M., Colot, V., Bossdorf, O., 2013. Epigenetic variation creates potential for evolution of plant phenotypic plasticity. *New Phytologist* 197.
- Zhang, S., Ghatak, A., Bazargani, M.M., Bajaj, P., Varshney, R.K., Chaturvedi, P., Jiang, D., Weckwerth, W., 2021. Spatial distribution of proteins and metabolites in developing wheat grain and their differential regulatory response during the grain filling process. *Plant Journal* 107.
- Zhang, S., Ghatak, A., Mohammadi Bazargani, M., Kramml, H., Zang, F., Gao, S., Ramsak, Z., Gruden, K., Varshney, R.K., Jiang, D., et al., 2024. Cell-type proteomic and metabolomic resolution of early and late grain filling stages of wheat endosperm. *Plant Biotechnol. J.* 22.
- Zhang, A., Liu, Y., Wang, F., Li, T., Chen, Z., Kong, D., Bi, J., Zhang, F., Luo, X., Wang, J., et al., 2019. Enhanced rice salinity tolerance via CRISPR/Cas9-targeted mutagenesis of the OsRR22 gene. *Molecular Breeding* 39.
- Zhang, Y., Xu, J., Li, R., Ge, Y., Li, Y., Li, R., 2023a. Plants' Response to Abiotic Stress: Mechanisms and Strategies. *Int. J. Mol. Sci.* 24.
- Zhang, M., Zhao, J., Hoshino, Y., 2023b. Deep learning-based high-throughput detection of in vitro germination to assess pollen viability from microscopic images. *J. Exp. Bot.* 74.
- Zhao, J., Bindi, M., Eitzinger, J., Ferrise, R., Gaile, Z., Gobin, A., Holzkämper, A., Kersebaum, K.C., Kozyra, J., Kriauciūnienė, Z., et al., 2022. Priority for climate adaptation measures in European crop production systems. *European Journal of Agronomy* 138.
- Zhao, C., Liu, B., Piao, S., Wang, X., Lobell, D.B., Huang, Y., Huang, M., Yao, Y., Bassu, S., Ciaia, P., et al., 2017. Temperature increase reduces global yields of major crops in four independent estimates. *Proceedings of the National Academy of Sciences*.
- Zhao, S., Zhang, Q., Liu, M., Zhou, H., Ma, C., Wang, P., 2021. Regulation of plant responses to salt stress. *Int. J. Mol. Sci.* 22.
- Zhong, X., Hong, W., Shu, Y., Li, J., Liu, L., Chen, X., Islam, F., Zhou, W., Tang, G., 2022. CRISPR/Cas9 mediated gene-editing of GmHdz4 transcription factor enhances drought tolerance in soybean (*Glycine max* [L.] Merr.). *Front. Plant Sci.* 13.
- Zhou, W., Chen, F., Meng, Y., Chandrasekaran, U., Luo, X., Yang, W., Shu, K., 2020. Plant waterlogging/flooding stress responses: From seed germination to maturation. *Plant Physiology and Biochemistry* 148.
- Zhu, T., Fonseca De Lima, C.F., De Smet, I., 2021. The heat is on: How crop growth, development, and yield respond to high temperature. *J. Exp. Bot.* 72.
- Zia, M.A.B., Ul-Allah, S., Sher, A., Ijaz, M., Sattar, A., Yousaf, M.F., Chaudhry, U.K., Qayyum, A., 2023. Genomics for Abiotic Stress Resistance in Legumes. *Sustainable Agriculture in the Era of the OMICs Revolution*.
- Zinn, K.E., Tunc-Ozdemir, M., Harper, J.F., 2010. Temperature stress and plant sexual reproduction: Uncovering the weakest links. *J. Exp. Bot.* 61, 1959–1968.
- Zioutopoulou, A., Patitaki, E., Xu, T., Kaiserli, E., 2021. The epigenetic mechanisms underlying thermomorphogenesis and heat stress responses in arabidopsis. *Plants* 10 (11), 2439.
- Zörb, C., Geilfus, C.M., Dietz, K.J., 2019. Salinity and crop yield. *Plant Biology* 21.
- Zou, H., Lu, H., Li, Y., Liu, L., Cao, Z., 2020. Maize tassels detection: A benchmark of the state of the art. *Plant Methods* 16.